Electron-Valley Resonance in monolayer Transition Metal Dichalcogenides

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ABSTRACT

The spin-valley resonance in the monolayer transition metal dichalcogenides is studied in this thesis. A certain material of 2D transition metal dichalcogenides mainly focused and investigated here is MoS_2. General electronic structure of the material is reviewed and with the method of Schrieffer-Wolff transformation a one-band effective model appears. By introducing quantum confinement and magnetic field we clarify the energy level structure of the quantum dot of monolayer MoS_2 as well as the Landau level physics. Through the special magnetic disorder mechanism, we argue and justify the realization of the spin-valley resonance in MoS_2 quantum dot system under magnetic field. The manipulation of the oscillation process and the quantum gate operation are discussed in the case of weak magnetic field.

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1 Introduction

Two dimensional semiconductor materials have been a hot discussed topic nowadays for their novel electronic and optic properties which are highly different from the normal three dimensional materials. Graphene as a first isolated atomic thick monolayer two dimensional material produced successfully by Andre Geim and Konstantin Novoselov *et al.*[1] has brought an great research interest on 2D materials for years. Graphene possesses highest electron mobility, electric- and thermal conductivity as well as IQHE observed under room temperature, which has great application advantage as a two dimensional material. However graphene has no band gap (hence the Dirac electron with zero effective mass is emergent at low energy) which makes it hard to be applied into the semiconductor materials field. Therefore people also turn to look at the other new members in 2D semiconductor family such as phosphorene and transition monolayer transition metal dichalcogenides (TMDCs) etc..

Phosphorene is like graphene, it is isolated usually from the bulk black phosphorus which is similar to graphite is a Van-der-Waals force connecting layered material. The high mobility under room temperature[2] and large gap along with good optical-electric property make phosphorene a potential candidate of new semiconductor device, e.g. constructing field-effect transistors.[3] The disadvantage of phosphrene is the unsuitability due to its oxidation in the open air environment.

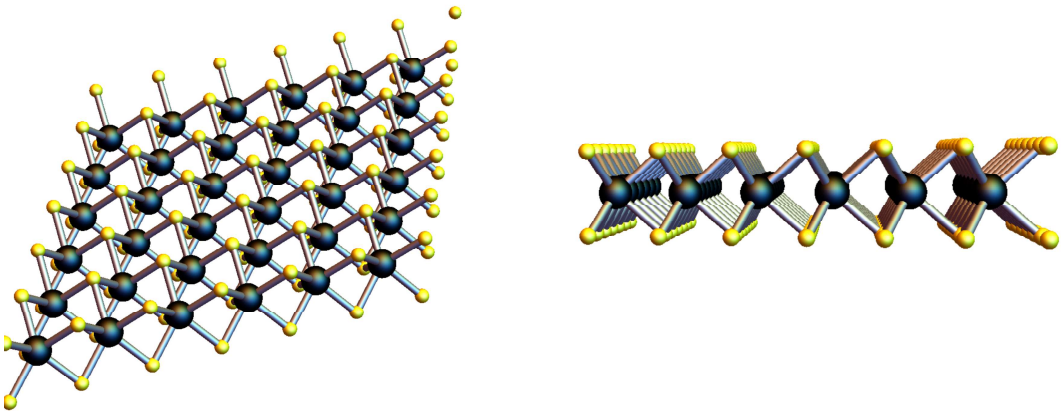
In these non-metal materials, spin-orbital coupling effect is relative weak for the atom has small mass. In transition metal dichalcogenides, the metal atoms are in VI element family with larger mass which enable stronger spin-orbital coupling to exist. The two dimensional TMDCs also known as monolayer TMDCs breaks the central inversion symmetry and this is the essential factor that realize the spin-valley splitting in the band structure. It was until 2010 the monolayer TMDCs was truly prepared by the group of Tony F. Heinz using molybdenum disulfide (MoS_2) crystal as original material.[4] Through theoretical studies and spectroscopic measurement, it has been shown the interplay between spin and valley physics in monolayer TMDCs brings non-trivial optical and electric transport process such as valley and spin Hall effects[5] and polarized excitations influenced by the spin- and valley optical transition selection rule which can be treated and understood with the Berry curvature kinetic theory.[5][6] In addition the mechanism of possible unconventional Ising superconductivity phase in TMDCs has also been investigated recently which will raise the crossing topic of novel 2D superconductor.[7]

In low dimensional semiconductor physics, quantum dot has always been frequently studied. Because of the quantum confinement effect, electrons restricted in such quasi-zero-dimension mesoscopic system exhibit discrete energy level structure. And the Coulomb blockade in the quantum dot often triggers many interesting phenomenon including the Fano effect in transport problem[8] and strongly correlation effect like Kondo effect.[9] These basic properties support the few (single) spin state in quantum dot system. By introducing spin-orbit and hyperfine interactions etc. the spin in quantum dot can be controlled and manipulated hence developing the fields of quantum information processing via spintronics with single spin in these mesoscopic systems.[10] The idea was then quickly extended to the valley degrees of freedom namely pseudospin in the 2D materials. For example of our interest here, in the well known carbon nanotube, Guido Burkard *et al.* has suggested a disorder mediated valley resonance mechanism which is able to carry out quantum gate operation on the quantum information process.[23] It is also very compelling to realize similar system in TMDCs with the spin-valley physics and carry out operation on spin as well as valley degrees of freedom. The quantum dot structure of TMDCs are theoretical and experimental studied and one can believe it is realistic as a semiconductor device.[14][11] The disorder in TMDCs from vacancies or impurities has also been investigated providing us an firm understanding on the disorder effect in TMDCs.[12] In the following text we present the full work based on these knowledge.

2 Lattice structure and electronic structure

2.1 Lattice structure of monolayer TMDCs

Monolayer Transition metal dichalcogenide (TMDC) materials can be viewed as a quasi-2D lattice in which the primitive unit-cell contain a MX_2, where M is the transition metal atom of VIB family in periodic table (Mo, W,etc.) and X is a chalcogen atom (S, Se, Te). The generic crystal lattice structure of monolayer TMDCs is shown in the following figures, which are plotted with the help of the program *Mathematica*. The code can be found in appendix.

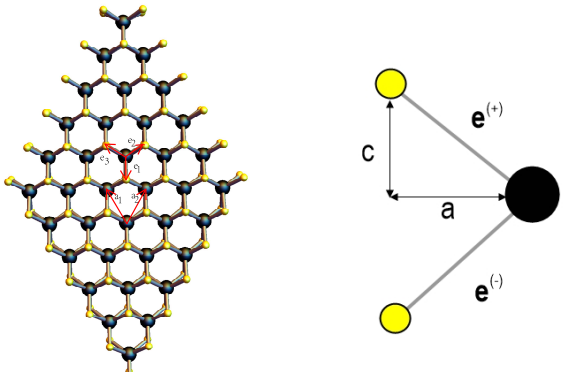


(a) (b)

Figure 1: Schematic lattice structure for monolayer TMDCs. Black balls represent VIB transition metal atoms M while yellow balls represent chalcogen atoms X.

The monolayer TMDCs are atomic thick and constructed with three atomic sublayers. One can see from above figures that the chalcogen atoms build up the upper and lower layer triangular sublattice and the transition metal atoms filling as the middle layer triangular sublattice whose orientation is reversal to those two from chalcogen atoms. Comparing to the well known graphene lattice which is a flat one-atom layer 2D lattice whose both two triangular sublattices are both built up from carbon atoms, the monolayer TMDC is a X-M-X sandwich structure with a layer sublattice from M and two layer sublattice from X, henceforth a quasi-2D lattice without sublattice symmetry.

Afterwards we come to discuss the detail lattice structure of monolayer TMDC with lattice constant. From the z-direction projection view, the lattice can be seen as a 2D honeycomb lattice, however it should be again reminded the X atoms in the triangular sublattice in this 2D projection picture contain upper- and lower layer part. Some basic distance constants are defined following the way



(a) (b)

Figure 2: Lattice constant and basis vectors in a unit cell of monolayer TMDCs lattice.

demonstrated in Fig.2 (b), $c = 1.51$Å is the vertical distance between a nearest pair of M and X atom whereas $a = 1.84$Å is the transverse distance between them.[13] In a primitive unit cell, we can choose the a M atom as original point, then there are six basis vectors pointing to the six X atoms from upper- and lower layer as sketched in Fig.2 (a). They are written as $\mathbf{e}_1^{(\pm)} = (0, -a, \pm c)$, $\mathbf{e}_2^{(\pm)} = (\frac{\sqrt{3}a}{2}, \frac{a}{2}, \pm c)$ and $\mathbf{e}_2^{(\pm)} = (-\frac{\sqrt{3}a}{2}, \frac{a}{2}, \pm c)$. The next nearest neighbor vectors (also six vectors), which connect the atom in the same layer (X-X or M-M) can be obtained from the combination of the six nearest neighbor vectors: $\mathbf{a}_1 = \mathbf{e}_3^{(\pm)} - \mathbf{e}_1^{(\pm)} = (-\frac{\sqrt{3}a}{2}, \frac{3a}{2}, 0)$, $\mathbf{a}_2 = \mathbf{e}_2^{(\pm)} - \mathbf{e}_1^{(\pm)} = (\frac{\sqrt{3}a}{2}, \frac{3a}{2}, 0)$ ect.

2.2 Tight-binding model

In order to establish the tight-binding model, we need to figure out the the atomic orbitals that contribute to the energy bands at the beginning. By the so far numerical computation studies, people believe the electronic structure of the TMDCs is mainly governed by the d-orbitals provided by M atoms and the p-orbitals from X atoms. By comparing the DFT numeric calculation with experiment results, Alexander J. Pearce *et al.*[13] have considered five localized d-orbitals from M atoms $d_{z^2}, d_{x^2-y^2}, d_{xy}, d_{xz}, d_{yz}$ and three basic localized p-orbitals from X atoms p_x, p_y, p_z. Together with these eleven orbitals in each primitive unit cell, the tight-binding model will generate a eleven bands electronic structure.

Due to the fact that the transition metal atom M has heavy atomic mass (comparing to carbon atom in graphene), in TMDCs we will also encounter large spin-orbital coupling effect contributing to the electronic structure. The tight-binding model then in principle should include the energy contribution from each local on-site atomic orbital energy, the electron hopping process between different atoms (same type as well as different type) and the spin-orbital coupling energy, hence the Hamiltonian is formally written as:

$$H = H_{\text{atom}} + H_{\text{hop}} + H_{\text{so}}. \tag{1}$$

In the second quantization language, we follow the notation in [13], operator $p^{\alpha}_{\mu,i,s}$ annihilates an electron on lattice site α =X1,X3 (X1 is the upper layer X atom while X3 is the lower layer X atom) in the $\mu = x, y, z$ orbital in the i-th primitive unit cell with spin $s =\uparrow, \downarrow$; $d^{\text{M2}}_{\nu,i,s}$ annihilates an electron on lattice site M2 (it means the M atom in the middle layer, and this index can be dropped out since there is only one layer of M atoms) in the orbitals $\nu = z^2, x^2-y^2, xy, xz, yz$ in the i-th primitive unit cell with spin $s =\uparrow, \downarrow$. According to the lattice structure, one can realize that in TMDCs the central

inversion symmetry is broken and the space symmetry group of the crystal breaks down to D_{3h} containing z-direction mirror symmetry σ_h.[14] Hence it is reasonable to recombine the chalcogen atom orbital operators into x-y plane mirror reflection ($z \to -z$) symmetric and antisymmetric sector, namely $p^{\mathrm{S}}_{\mu,i,s} = \frac{1}{\sqrt{2}}(p^{\mathrm{X1}}_{\mu,i,s} + p^{\mathrm{X3}}_{\mu,i,s})$ and $p^{\mathrm{A}}_{\mu,i,s} = \frac{1}{\sqrt{2}}(p^{\mathrm{X1}}_{\mu,i,s} - p^{\mathrm{X3}}_{\mu,i,s})$. One can further combine them into rotating orbital basis so that angular momentum can be more transparent: for p-orbitals $p^{\beta}_{\pm1,i,s} = \frac{1}{\sqrt{2}}(p^{\beta}_{x,i,s} \pm i p^{\beta}_{y,i,s})$ where $\beta = \mathrm{S,A}$; for d-orbitals $d_{\pm2} = \frac{1}{\sqrt{2}}(d_{x^2-y^2,i,s} \pm i d_{xy,i,s})$, $d_{\pm1,i,s} = \frac{1}{\sqrt{2}}(d_{xy,i,s} \pm i d_{yz,i,s})$. The index $\pm1, \pm2$ refers to the z-component angular momentum quantum number of the states in the rotating orbital basis. With these notation, the on-site atomic orbitals energies H_{atom} and hopping energies $H_{\mathrm{hop}} = H_{dd} + H_{dp} + H_{pd}$ can be expressed with on-site energy parameters $\varepsilon^{l}_{p,d}$ ($l = 1, 2$ is the angular momentum quantum number) and hopping energy parameters $t^{ij}_{\mu\nu}$. As for the spin-orbital coupling, people usually start with fundamental expression $H_{\mathrm{so}} = \frac{\hbar}{4m_e^2c^2}\frac{1}{r}\frac{dV(r)}{dr}\boldsymbol{L}\cdot\boldsymbol{S}$ where $V(r)$ is the atom potential, $\boldsymbol{L}$ and $\boldsymbol{S}$ are orbital angular momentum operator and spin operator respectively. The derivation on this term can be found in any quantum mechanics book. By evaluating the mean value of the spin-orbital couping energy with wave function of the M,X atoms, the spin-orbital coupling constant can be introduce to the tight-binding model

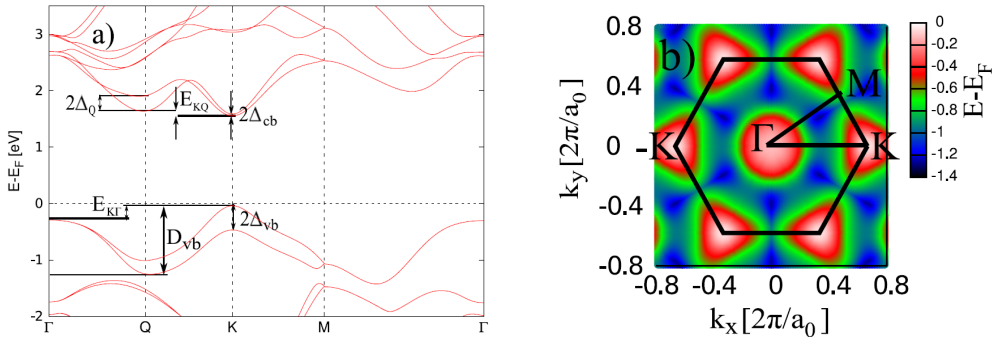
$$\lambda_{\mathrm{M}} = \frac{\hbar}{4m_e^2c^2}\int d\boldsymbol{r} R^{*}_{\mathrm{M},l}(r)\frac{1}{r}\frac{dV(r)}{dr}R_{\mathrm{M},l}(r) \tag{2}$$

$$\lambda_{\mathrm{X}} = \frac{\hbar}{4m_e^2c^2}\int d\boldsymbol{r} R^{*}_{\mathrm{X},l}(r)\frac{1}{r}\frac{dV(r)}{dr}R_{\mathrm{X},l}(r), \tag{3}$$

where $R_{\mathrm{M(X)},l}(r)$ is the radial part wave function of M,X atom. The atomic potential is assumed to be sphereical symmetric. The full detail expression of the tight-binding model Hamiltonian will not be presented in here, one in interest can refer to [13].

2.3 Energy bands near K point

The DFT calculation on the band structure in Brillouin zone (BZ) shows the maximum of valence band and minimum of conduction band are at two-fold degenerate K points, we will label them as K and K′ through out this work.



(a) Band structure along Γ-Q-K-M-Γ

(b) Band structure in 2D BZ

Figure 3: Feature band structure of monolayer TMDCs in BZ from DFT calculations. E_F is the Fermi energy whereas $\Gamma = (0,0)$, Q= $(\frac{\pi}{2a_0}, 0)$, K= $(\frac{4\pi}{3a_0}, 0)$, M= $(\frac{\pi}{a_0}, \frac{\pi}{\sqrt{3}a_0})$ are the high symmetry points in Brillouin zone ($a_0 = 2a\cos 30°$). The figures are taken from [16].

The wave vector of K points are according to the lattice constant given as $(\tau\frac{4\pi}{3\sqrt{3}a}, 0)$ where the valley index τ are introduced for K ($\tau = +1$) and K′ ($\tau = -1$). To describe the bands theoretically, we

need to transform our Hamiltonian from real lattice space to **k**-space. The Fourier transformation on the operators are written as:

$$p^{\beta}_{\mu,i,s} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} p^{\beta}_{\mu,\boldsymbol{k},s} \tag{4}$$

$$d_{\nu,i,s} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} d_{\nu,\boldsymbol{k},s} \tag{5}$$

With the help of the basic tight binding model, one can obtain Hamiltonian in momentum space $H = \sum_{\boldsymbol{k}} \psi^{\dagger}_{\boldsymbol{k}} H_{\boldsymbol{k}} \psi_{\boldsymbol{k}}$ describing the eleven electronic bands. Dividing the total Hilbert space into two subspaces according to the even and odd symmetries with respect to mirror inversion of the middle transition metal layer $\psi_{\boldsymbol{k}} = (\psi^{\mathrm{E}}_{\boldsymbol{k}}, \psi^{\mathrm{O}}_{\boldsymbol{k}})^T$ where $\psi^{\mathrm{E}}_{\boldsymbol{k}} = (d_{z^2,\boldsymbol{k},s}, d_{\tau 2,\boldsymbol{k},s}, d_{-\tau 2,\boldsymbol{k},s}, p^{\mathrm{S}}_{\tau 1,\boldsymbol{k},s}, p^{\mathrm{S}}_{-\tau 1,\boldsymbol{k},s}, p^{\mathrm{A}}_{z,\boldsymbol{k},s})^T$ and $\psi^{\mathrm{O}}_{\boldsymbol{k}} = (d_{\tau 1,\boldsymbol{k},s}, d_{-\tau 1,\boldsymbol{k},s}, p^{\mathrm{A}}_{\tau 1,\boldsymbol{k},s}, p^{\mathrm{A}}_{-\tau 1,\boldsymbol{k},s}, p^{\mathrm{S}}_{z,\boldsymbol{k},s})$, the Hamiltonian yields a even-odd block form[13]:

$$H_{\boldsymbol{k}} = \begin{pmatrix} H^{\mathrm{E}} & H^{\mathrm{EO}} \\ H^{\mathrm{OE}} & H^{\mathrm{O}} \end{pmatrix}, \; H^{\mathrm{E}} = \begin{pmatrix} H^{\mathrm{E}}_{\boldsymbol{k},dd} & H^{\mathrm{E}}_{\boldsymbol{k},dp} \\ H^{\mathrm{E}}_{\boldsymbol{k},pd} & H^{\mathrm{E}}_{\boldsymbol{k},pp} \end{pmatrix}, \; H^{\mathrm{O}} = \begin{pmatrix} H^{\mathrm{O}}_{\boldsymbol{k},dd} & H^{\mathrm{O}}_{\boldsymbol{k},dp} \\ H^{\mathrm{O}}_{\boldsymbol{k},pd} & H^{\mathrm{O}}_{\boldsymbol{k},pp} \end{pmatrix}, \tag{6}$$

The off-diagonal block matrices above originate from the spin-orbital coupling, which mix up the even and odd sectors. We want to take a closer look at the low energy behavior $\mathbf{K}(\mathbf{K}')$ point. Around this symmetric point, one can known from the numeric results based on DFT calculations shown in Fig.4 that the lowest conduction- and highest valence bands all possess an even symmetry.

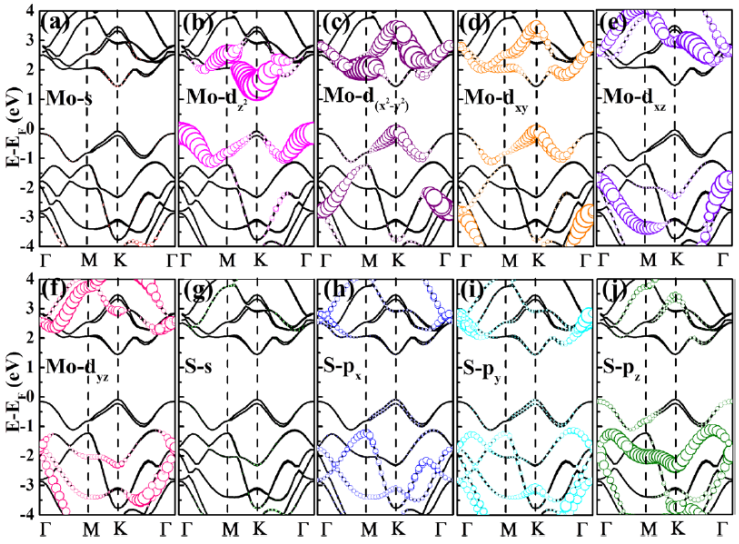


Figure 4: Band structures of the monolayer MoS_2 calculated by DFT with demonstration on the separated contributions from various orbital components labeled directly in each subfigures. The size of the circle in the figure resembles the intensity of contribution from the relevant orbital component. The whole figure is taken from [15].

Therefore we focus on the even sector bands around **K** points. By expanding the dispersion around $\mathbf{K} = \left(\frac{4\pi a}{3\sqrt{3}}, 0\right)$ $(\mathbf{K}' = -\mathbf{K})$ regrading to small deviation momentum at each valley point $\boldsymbol{k} = \pm\mathbf{K} + \boldsymbol{q}$. With definition of valley symbol $\tau = \pm 1$ and recombining components of momentum vector into $q^{\pm}_{\tau} = \tau q_x \pm i q_y$, one can rewrite the Hamiltonian with $q^{\pm}_{\tau}$ as variables. Detail expression of the full Hamiltonian with large amount of parameters is again referred to [13].

3 Effective theory at low energy

3.1 Two-band model

Meanwhile it can be noticed that p-bands are higher energy part comparing to d-bands, hence we aim to construct low-energy effective Hamiltonian regarding to the lowest energetic d-bands.The tunneling between d- and p bands flips the energy states causing mixing and splitting of these bands. The even sector is now divided according to energies as

$$H_{\boldsymbol{q}}^{E} = \begin{pmatrix} H_{\mathrm{L}} & V^{\dagger} \\ V & H_{\mathrm{H}} \end{pmatrix}, \tag{7}$$

where the low energy part consists two lowest-energy d-bands from $(d_{\tau 2,\boldsymbol{k},s}, d_{z^2,\boldsymbol{k},s})$ while higher energy part contain another one d-band and three p-bands, i.e. $(d_{-\tau 2,\boldsymbol{k},s}, p^{\mathrm{S}}_{\tau 1,\boldsymbol{k},s}, p^{\mathrm{S}}_{-\tau 1,\boldsymbol{k},s}, p^{\mathrm{A}}_{z,\boldsymbol{k},s})$. By using the method of Schrieffer-Wolf canonical transformation, one can project the this six-band Hilbert space from even sector into two lowest energetic d-bands (lowest conductance band and highest valence band) expanding subspace and obtain an effective Hamiltonian[13]:

$$H^{\tau}_{\mathrm{eff}} = H^{\tau}_{0} + H^{\tau}_{\mathrm{trig}} + H^{\tau}_{\mathrm{cub}} + H_{\mathrm{so}}, \tag{8}$$

$$H_{0} = \begin{pmatrix} \varepsilon_{\mathrm{vb}} & \gamma q^{+}_{\tau} \\ \gamma q^{-}_{\tau} & \varepsilon_{\mathrm{cb}} \end{pmatrix} + \begin{pmatrix} \alpha |q_{\tau}|^{2} & 0 \\ 0 & \beta |q_{\tau}|^{2} \end{pmatrix}, \tag{9}$$

$$H^{\tau}_{\mathrm{trig}} = \kappa \begin{pmatrix} 0 & (q^{+}_{\tau})^{2} \\ (q^{-}_{\tau})^{2} & 0 \end{pmatrix}, \tag{10}$$

$$H^{\tau}_{\mathrm{cub}} = |q_{\tau}|^{2} \begin{pmatrix} \tau\omega_{\mathrm{cb}} |q_{\tau}| \cos(3\varphi_{q}) & \eta q^{+}_{\tau} \\ \eta q^{-}_{\tau} & \tau\omega_{\mathrm{vb}} |q_{\tau}| \cos(3\varphi_{q}) \end{pmatrix}, \tag{11}$$

$$H_{\mathrm{so}} = \begin{pmatrix} \Delta_{\mathrm{vb}} \tau s_{z} & 0 \\ 0 & \Delta_{\mathrm{cb}} \tau s_{z} \end{pmatrix}. \tag{12}$$

In above we have adopted the valley dependent wave vector $q^{\pm}_{\tau} = \tau q_{x} \pm i q_{y}$. $\varepsilon_{\mathrm{vb(cb)}}$ represents the valence (conduction) band edge energy. α and β can be understood as the effective mass term of the corresponding band. H^{τ}_{trig} produces trigonal warping, H^{τ}_{cub} is a cubic momentum term which is responsible for the away from K, K' point band behavior.[17] The intrinsic SOC term $H^{\mathrm{in}}_{\mathrm{so}} = \Delta_{\mathrm{cb}} \tau s_{z}$ causes spin-split of conduction band (cb). All the parameters here can be expressed from the parameters in the previous tight-biding model. In some situation by applying vertical external electric field breaking the inversion symmetry, a Rashba SOC will also appear,

$$H_{\mathrm{R}} = \lambda^{*}_{\mathrm{R}} q^{+}_{\tau} s_{-} + \lambda_{\mathrm{R}} q^{-}_{\tau} s_{+}. \tag{13}$$

The Rashba SOC is relatively small therefore one can handle it as a perturbation term, whereas the stronger intrinsic SOI is treated exactly.[14]

3.2 One-band model

One can further project the Hamiltonian to one band effective form again with the Schrieffer-Wolff transformation. Decomposing the two-band model again according to the low- and high energy subspace, Hamiltonian then reads formally $H = H_{0} + H_{I}$ where the H_{0} is diagonal free part of low- and high energy sector while the H_{I} is the off-diagonal interaction part mixing these to sectors up, in matrix form it is the same like (7). The spirit of Schrieffer-Wolff transformation is that one

can carry out a unitary transformation that maps the previous Hamiltonian effectively to block diagonal form without off-diagonal elements[18]:

$$U\begin{pmatrix} H_L & V^\dagger \\ V & H_H \end{pmatrix}U^\dagger = \begin{pmatrix} H^* & 0 \\ 0 & H' \end{pmatrix}, \tag{14}$$

where the unitary operator is normally defined as $U = e^S$ hence the S operator is anti-unitary $S^\dagger = -S$. With the help of BCH (Baker-Campbell-Hausdorff) formula, one can expand the transformation as $\tilde{H} = e^S(H_0 + H_I)e^{-S} = H + [S,H] + \frac{1}{2!}[S,[S,H]] + \frac{1}{3!}[S,[S,[S,H]]] + \cdots = H_0 + H_I + [S,H_0] + [S,H_I] + \frac{1}{2!}[S,[S,H]] + \cdots$, and by setting

$$[S,H_0] = -H_I, \tag{15}$$

all the off-diagonal terms can be eliminated and the we arrive at:

$$\tilde{H} = H_0 + \frac{1}{2}[S,H_I] + \sum_{n=2}\frac{n}{(n+1)!}[S,H_I]_n = H_0 + \Delta H. \tag{16}$$

To the lowest order perturbation we obtain $\Delta H = \frac{1}{2}[S,H_I]$. Now we aim to obtain the effective Hamiltonian for low energy sector, after these perturbation, we can simply use the projector $P_L = \sum_{\{L\}}|L\rangle\langle L|$ to project total Hamiltonian into effective low-energy part $H^* = H_L + \Delta H_L$, $\Delta H_L = \frac{1}{2}P_L[S,H_I]P_L$. In the matrix form the S operator can be written as

$$S = \begin{pmatrix} 0 & -s^\dagger \\ s & 0 \end{pmatrix}. \tag{17}$$

When $(H_L)_{ab} = E_{L,a}\delta_{ab}$ and $(H_H)_{ab} = E_{H,a}\delta_{ab}$, by applying (15) the elements in S can be derived as $s_{ab} = \frac{V_{ab}}{E_{H,a}-E_{L,b}}$ and $-s^\dagger = \frac{V^\dagger_{ab}}{E_{L,a}-E_{H,b}}$. Therefore the effective corrected Hamiltonian for low-energy is then given by:

$$\Delta H_{LL'} = -\frac{1}{2}P_L(V^\dagger s + s^\dagger V)P_{L'} = -\frac{1}{2}\sum_{\{H\}}\Big[\frac{V^\dagger_{LH}V_{HL'}}{E_H - E_L} + \frac{V^\dagger_{LH}V_{HL'}}{E_H - E_{L'}}\Big]. \tag{18}$$

From the two-band model Hamiltonian, we can see the high energy sector describe the conduction band while valence band is the low energy sector. Hence we have $E_L = \varepsilon_{\rm vb} + \alpha|q_\tau|^2$ and $E_H = \varepsilon_{\rm cb} + \beta|q_\tau|^2$. We keep only the quadratic momentum term in the effective Hamiltonian, therefore we drop out the high order of momentum term and set $V^\dagger = \gamma q_\tau^+$, $V = \gamma q_\tau^-$. Then substitute these elements into the lowest order corrected Hamiltonian above and get:

$$\Delta H_L = \frac{\gamma^2 q_\tau^+ q_\tau^-}{(\varepsilon_{\rm vb} - \varepsilon_{\rm cb}) + (\alpha - \beta)|q_\tau|^2} + \Delta_{\rm vb}\tau s_z \approx \frac{\gamma^2 q_\tau^+ q_\tau^-}{\varepsilon_{\rm vb} - \varepsilon_{\rm cb}} + \Delta_{\rm vb}\tau s_z. \tag{19}$$

Similarly we can project out the valence band $\Delta H_{HH'} = \frac{1}{2}P_H(sV^\dagger + Vs^\dagger)P_{H'}$ and obtain the effective Hamiltonian for conduction band:

$$H^{\tau,s}_{\rm cb} = H_H + \Delta H_H \approx \beta|q_\tau|^2 + \frac{\gamma^2 q_\tau^- q_\tau^+}{\varepsilon_{\rm cb} - \varepsilon_{\rm vb}} + \Delta_{\rm cb}\tau s_z. \tag{20}$$

This result is compatible with those obtained from $\mathbf{k}\cdot\mathbf{p}$ perturbation method.[14][16]

4 Quantum dot model in TMDC under magnetic field

4.1 System under magnetic field

In the presence of an external magnetic field, the canonical wave vector $\boldsymbol{q}$ should be replaced by mechanical wave vector $\boldsymbol{\pi} = \boldsymbol{q} + \frac{e}{\hbar}\boldsymbol{A}$ in the Hamiltonian. Its components $\pi^{\pm}$ no longer commute with each other:

$$[\pi^-, \pi^+] = [\pi_x - i\pi_y, \pi_x + i\pi_y] = i[\pi_x, \pi_y] - i[\pi_y, \pi_x] = 2i[\pi_x, \pi_y] = \frac{2e}{\hbar}B_z. \tag{21}$$

In above derivation we have used the commutator $[\pi_i, \pi_j] = -\frac{ie}{\hbar}\epsilon^{ijk}\partial_i A_j = -\frac{ie}{\hbar}B_k$ by employing Einstein's summation convention. Then under the valley index notation, the communicator is simply modified to $[\pi_\tau^-, \pi_\tau^+] = \frac{2e}{\hbar}\tau B_z$. The effective conduction band Hamiltonian is then changed to:

$$\begin{aligned} H_{\mathrm{cb}}^{\tau,s} &= \beta\frac{\pi_\tau^+\pi_\tau^- + \pi_\tau^-\pi_\tau^+}{2} + \frac{\gamma^2\pi_\tau^-\pi_\tau^+}{\varepsilon_{\mathrm{cb}} - \varepsilon_{\mathrm{vb}}} + \Delta_{\mathrm{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z \\ &= \frac{\hbar^2\pi_\tau^+\pi_\tau^-}{2m_{\mathrm{eff}}} + \Delta_{\mathrm{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z + \frac{1}{2}g_v\mu_B\tau B_z, \end{aligned} \tag{22}$$

where the new parameters are interpreted as effective electron mass $\frac{1}{m_{\mathrm{eff}}} = \frac{2}{\hbar^2}\left(\beta + \frac{\gamma^2}{\varepsilon_{\mathrm{cb}}-\varepsilon_{\mathrm{vb}}}\right)$ and effective electron valley g-factor $g_v = \frac{2e}{\hbar}\mu_B^{-1}\left(\beta + \frac{2\gamma^2}{\varepsilon_{\mathrm{cb}}-\varepsilon_{\mathrm{vb}}}\right) = \frac{4m_e}{\hbar^2}\left(\beta + \frac{2\gamma^2}{\varepsilon_{\mathrm{cb}}-\varepsilon_{\mathrm{vb}}}\right)$, Bohr magneton $\mu_B = \frac{e\hbar}{2m_e}$. At first the effective mass of electron in MoS_2 is taken as $m_{\mathrm{eff}} = 0.47m_e$ where we do average on the two kind of spin-valley electron effective mass value[14]. The second term is intrinsic spin-orbital coupling describing the spin splits of the bands where the coupling constant is numerically evaluated as $\Delta_{\mathrm{cb}} = 1.5\mathrm{meV}$ via DFT approach, whereas the third term in (22) is known as electron Zeeman splitting energy under magnetic field and the electron g-factor g_e is approximately 2 while the last term are emergent valley Zeeman energy which splits the two valley bands K, K' in energy level. The effective valley g-factor is chosen as $g_v = 3.57$ referring the numeric results also based on $\mathbf{k}\cdot\mathbf{p}$ perturbation and DFT performed by Andor Kormányos *et al.* .[14]

Within first quantization framework, mechanical wave vector yields $\pi_\tau^{\pm} = \tau(-i\partial_x + \frac{e}{\hbar}A_x) \pm i(-i\partial_y + \frac{e}{\hbar}A_y) = -i(\tau\partial_x \pm i\partial_y) + \frac{e}{\hbar}(\tau A_x \pm iA_y)$. In polar coordinate (r, φ) the partial derivatives convert to $\partial_x = \cos\varphi\partial_r - \frac{1}{r}\sin\varphi\partial_\varphi$, $\partial_y = \sin\varphi\partial_r + \frac{1}{r}\cos\varphi\partial_\varphi$ and under the symmetric gauge $A_x = -\frac{1}{2}B_z y$, $A_y = \frac{1}{2}B_z x$, by introducing the magnetic length $l_B^2 = \frac{\hbar}{eB_z}$ and redefining a new dimensionless radial variable $\rho = \frac{1}{2}(\frac{r}{l_B})^2 \approx \frac{B_z}{2}(\frac{r}{25\mathrm{nm}})^2$, two mechanical wave vector components take the following operator form:

$$\begin{aligned} \pi_\tau^{\pm} &= \tau(k_x + \frac{e}{\hbar}A_x) \pm i(k_y + \frac{e}{\hbar}A_y) = (\tau k_x \pm ik_y) + \frac{e}{\hbar}(\tau A_x \pm iA_y) \\ &= -i[\tau(\cos\varphi\partial_r - \frac{1}{r}\sin\varphi\partial_\varphi) \pm i(\sin\varphi\partial_r + \frac{1}{r}\cos\varphi\partial_\varphi)] + \frac{eB}{2\hbar}(-\tau\, r\sin\varphi \pm ir\cos\varphi) \\ &= -i(\tau\cos\varphi \pm i\sin\varphi)\partial_r - i(-\tau\sin\varphi \pm i\cos\varphi)\frac{\partial_\varphi}{r} + \frac{eB}{2\hbar}r(-\tau\sin\varphi \pm i\cos\varphi) \\ &= -i\tau e^{\pm i\tau\varphi}\partial_r \pm e^{\pm i\tau\varphi}\frac{\partial_\varphi}{r} \pm i\frac{eB}{2\hbar}re^{\pm i\tau\varphi} \\ &= \pm ie^{\pm i\tau\varphi}(\frac{eB}{2\hbar}r \mp \tau\partial_r - i\frac{1}{r}\partial_\varphi) \\ &= \frac{\pm i}{l_B}\sqrt{\frac{\rho}{2}}e^{\pm i\tau\varphi}(1 \mp 2\tau\partial_\rho - \frac{i}{\rho}\partial_\varphi) = \frac{\sqrt{2}}{l_B}a_\tau^{\pm}. \end{aligned}$$

This directly gives rise to the Landau level Hamiltonian of electrons in TMDC under magnetic field:

$$H = \hbar\omega_c(a_\tau^\dagger a_\tau + \frac{1}{2}) + \Delta_{\text{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z + \frac{1}{2}g_v\mu_B\tau B_z, \tag{23}$$

with $\omega_c = \frac{eB_z}{m_{\text{eff}}}$ as cyclotron frequency. The aspect of Landau level will be discussed later. In following part we want to bring in the investigation on quantum dot structure which is the most important structure in this work.

4.2 Quantum dot structure

Now consider a quantum dot (QD) structure in a small circular shpaed region of two-dimensional TMDCs materials, which is usually constructed by applying an external electrostatic potential with gates.[14] Here in this article we mainly focus on monolayer MoS_2, through the gates-voltage applied on the material sheet, we can manifest small well that forms a circular QD structure with a hard-wall potential: $U_{\text{dot}}(r) = U_0\Theta(r - R_d)$, where $U_0 \gg 1$ (one can also take infinity high barrier limit $U_0 \to \infty$ here) and $\Theta(x)$ is the Heaviside step function. Such sort of structure with the quantum confinement effect was latest claimed to be realized in the experiment.[11]

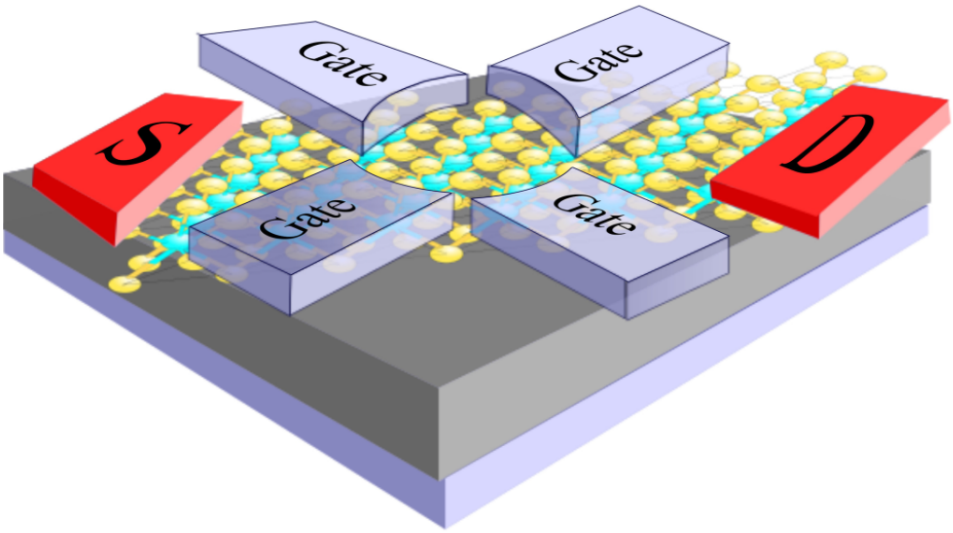


Figure 5: Schematic circular quantum dot structure produced from controlling voltage gates. The figure is taken from [14].

Here we take $B_z > 0$ without loss of generality. The Schrödinger equation is then derived as $\left[\hbar\omega_c(a_\tau^\dagger a_\tau + \frac{1}{2}) + H^{\tau,s} + U(\rho)\right]\Psi^{\tau,s}(\rho,\varphi) = E\Psi^{\tau,s}(\rho,\varphi)$ with $H^{\tau,s} = \Delta_{\text{cb}}\tau s_z + \frac{1}{2}g_e\mu_B s_z B_z + \frac{1}{2}g_v\mu_B\tau B_z$, $U(\rho) = U_0\Theta(\rho - \rho_d)$, $\rho_d = \rho(R_d)$ and $\hbar\omega_c = \frac{e|B_z|}{m_{\text{eff}}}$ is the cyclotron frequency. The eigenfunctions of operator $a^\dagger a$ need to be resolved. Notice that QD system has a cylindrical symmetry and therefore the z-component of angular momentum operator $L_z = -i\hbar\partial_\varphi$ commutes with Hamiltonian and shares common eigenfunctions. Due to this fact we can factorize the eigenfunction as $\Psi_m^{\tau,s}(\rho,\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}\Phi_m^{\tau,s}(\rho)\chi_s$ where $m \in \mathbb{Z}$. Now we write out $a_\tau^\dagger a_\tau$ in operator form explicitly:

$$\begin{aligned} a_\tau^\dagger a_\tau &= \frac{\sqrt{\rho}}{2}e^{+i\tau\varphi}\Big(1 - 2\tau\partial_\rho - \frac{i}{\rho}\partial_\varphi\Big)\frac{\sqrt{\rho}}{2}e^{-i\tau\varphi}\Big(1 + 2\tau\partial_\rho - \frac{i}{\rho}\partial_\varphi\Big) \\ &= \frac{\rho}{4} - \frac{\tau}{2} - \frac{i}{2}\partial_\varphi - \frac{1}{4\rho}\partial_\varphi^2 - \partial_\rho - \rho\partial_\rho^2. \end{aligned}$$

Then we substitute $\Phi_m^{\tau,s}(\rho,\varphi)$ into eigenequation and then analyze asymptotic behaviors at $\rho\to+\infty$ as well as $\rho\to 0$. When $\rho\to+\infty$, differential equation becomes $\frac{1}{4}\Phi_m^{\tau,s}(\rho)-\frac{\partial^2\Phi_m^{\tau,s}(\rho)}{\partial\rho^2}=0$ and can be solved with asymptotic form $\Phi_m^{\tau,s}(\rho)\sim e^{-\rho/2}$ (here $e^{+\rho/2}$ is abandoned due to its divergence at $\rho\to\infty$). On th other hand, $\rho\to 0$ equation tends to $[-\frac{1}{4\rho^2}\partial_\varphi^2-\frac{1}{\rho}\partial_\rho-\partial_\rho^2]e^{im\varphi}\Phi_m^{\tau,s}(\rho)=0$, and we assume $\Phi_m^{\tau,s}(\rho)\overset{\rho\to 0}{\sim}\rho^\xi$. Null condition yields $\frac{m^2}{4}-\xi^2=0$ which is solved as $\xi=\pm\frac{m}{2}$. Wave function should be convergent at $\rho=0$ hence $\xi=\frac{|m|}{2}$ should be choosed. Then the radial part of wave function should take a more detail form $\Phi_m^{\tau,s}(\rho)=\rho^{|m|/2}e^{-\rho/2}u_m^{\tau,s}(\rho)$. Now we apply the total $a_\tau^\dagger a_\tau$ operator to the trial wave function: $\Psi_m^{\tau,s}(\rho,\varphi)$:

$$\begin{aligned}a_\tau^\dagger a_\tau\Psi_m^{\tau,s}(\rho,\varphi)&=\frac{1}{4}e^{im\varphi}e^{-\rho/2}\rho^{\frac{|m|}{2}-1}\\&\quad\times\left[2\rho(m+|m|+1-\tau)u_m^{\tau,s}(\rho)-4\rho(|m|+1-\rho)\frac{d}{d\rho}u_m^{\tau,s}(\rho)-4\rho^2\frac{d^2}{d\rho^2}u_m^{\tau,s}(\rho)\right]\\&=e^{im\varphi}e^{-\rho/2}\rho^{\frac{|m|}{2}}\left[\frac{1}{2}(m+|m|+1-\tau)u_m^{\tau,s}(\rho)-(|m|+1-\rho)\frac{d}{d\rho}u_m^{\tau,s}(\rho)-\rho\frac{d^2}{d\rho^2}u_m^{\tau,s}(\rho)\right].\end{aligned}$$

According to the Schrödinger equation we formally obtain the following second-order ordinary differential equation:

$$\rho\frac{d^2u_m^{\tau,s}(\rho)}{d\rho^2}+(|m|+1-\rho)\frac{du_m^{\tau,s}(\rho)}{d\rho}-a_m^{\lessgtr}u_m^{\tau,s}(\rho)=0,\tag{24}$$

which is a confluent hypergeometric differential equation, also known as Kummer's differential equation. It has a regular singular point at 0 and an irregular singularity at infinity. The parameters $a_m^{\lessgtr}$ for the region $\rho<\rho_d$ and $\rho>\rho_d$ are derived as:

$$\hbar\omega_c a_m^{<}=\frac{1}{2}(m+|m|+1-\tau)\hbar\omega_c+H^{\tau,s}-E\tag{25}$$

$$\hbar\omega_c a_m^{>}=\hbar\omega_c a_m^{<}+U_0\tag{26}$$

In general, the universal solution of Kummer's differential equation can be expressed with the first and second kind of confluent hypergeometric function, i.e. $C_1M(a,|m|+1,\rho)+C_2U(a,|m|+1,\rho)$. However we need to figure put solution with reasonable physical meaning, which gives certain constrain on the combination of these two kinds of function. The wave function should in principle satisfy some basic properties: single-valued, convergent (namely remaining finite in all region and approaches to zero at infinity), smooth ($u_m^{\tau,s}(\rho)\in C^1[0,+\infty]$) and normalized. Hence we can simply observe the asymptotic behavior of these confluent hypergeometric functions. Following the the properties of these functions which can be learned from [19] in detail, we know that the first kind follows $\lim_{\rho\to0}M(a,|m|+1,\rho)=1$, $\lim_{\rho\to+\infty}M(a,|m|+1,\rho)=\infty$ while the second kind $\lim_{\rho\to0}U(a,|m|+1,\rho)\simeq\lim_{\rho\to0}\frac{\Gamma(|m|)}{\Gamma(a)}\rho^{-|m|}=\infty$, $\lim_{\rho\to+\infty}U(a,|m|+1,\rho)\simeq\lim_{\rho\to+\infty}\rho^{-a}$. These behaviors imply that in the inner region within QD the second kind confluent hypergeometric function should not appear while for the convergence in outer region outside the QD the first kind should vanish. Therefore we may attempt to construct the solution with the help of step function to cut the radius wave function into piecewise function:

$$u_m^{\tau,s}(\rho)=C^{<}M(a_m^{<},|m|+1,\rho)\Theta(\rho_d-\rho)+C^{>}U(a_m^{>},|m|+1,\rho)\Theta(\rho-\rho_d),\tag{27}$$

$C^{<}$ and $C^{>}$ are the normalization coefficients for inner- and outer region respectively. Then we turn back to investigate the infinite far behavior of $U(a_m^{>},|m|+1,\rho)\overset{\rho\to+\infty}{\sim}\rho^{-a_m^{>}}$. The parameter $a_m^{<}$ could be negative, however since the hard-wall potential U_0 is much larger then any other value of parameters, we can simply argue that $a_m^{>}\gg1$ and then $\lim_{\rho\to+\infty}\rho^{-a_m^{>}}=0$ meets the requirement of wave function property. There is still one thing is missing, that is the continuousness. At the

boundary of QD $\rho = \rho_d$, these two hypergeometric functions should be able to smoothly connect with each other, i.e. $C^{<}M(a_m^{<}, |m|+1, \rho_d) = C^{>}U(a_m^{>}, |m|+1, \rho_d)$, $C^{>}\partial_\rho M(a_m^{<}, |m|+1, \rho)|_{\rho_d} = C^{<}\partial_\rho U(a_m^{>}, |m|+1, \rho)|_{\rho_d}$. The derivative can be calculated out, one can find the smooth connection condition is a linear equation system for $C^{\lessgtr}$:

$$\begin{bmatrix} M(a_m^{<}, |m|+1, \rho_d) & U(a_m^{>}, |m|+1, \rho_d) \\ \frac{a_m^{<}}{|m|+1} M(a_m^{<}+1, |m|+2, \rho_d) & -a_m^{>} U(a_m^{>}+1, |m|+2, \rho_d) \end{bmatrix} \begin{bmatrix} C^{<} \\ C^{>} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{28}$$

The existence of non-zero solutions for $C^{\lessgtr}$ demands the Cramer's rule, hence we need to calculate the the determinant of the coefficient matrix, wich is actually Wronskian between these two confluent hypergeometric functions $W\{M(a_m^{<}, |m|+1, \rho_d), U(a_m^{>}, |m|+1, \rho_d)\}$:

$$\det \begin{bmatrix} M(a_m^{<}, |m|+1, \rho_d) & U(a_m^{>}, |m|+1, \rho_d) \\ \frac{a_m^{<}}{|m|+1} M(a_m^{<}+1, |m|+2, \rho_d) & -a_m^{>} U(a_m^{>}+1, |m|+2, \rho_d) \end{bmatrix} = 0, \tag{29}$$

namely $a_m^{>} M(a_m^{<}, |m|+1, \rho_d) U(a_m^{>}+1, |m|+2, \rho_d) + \frac{a_m^{<}}{|m|+1} M(a_m^{<}+1, |m|+2, \rho_d) U(a_m^{>}, |m|+1, \rho_d) = 0$. By setting the QD radius $R_d = 50\text{nm}$, ρ_d can directly rewritten with magnetic field as $(\frac{50\text{nm}}{25\text{nm}})^2 \frac{B_z}{2} = 2B_z$. This is a transcendental equation and through employing the program *Mathematica*, we can solve it with respect to $a_m^{\lessgtr}$ as the variable along with the continuous altering parameter B_z. And during solving process one can find that with every certain parameters setting (fixed B_z and m) there are multiple roots of $a_m^{\lessgtr}$ and in order to label different roots we introduce a new index $n \in \mathbb{N}$ for $a_m^{\lessgtr}$ turning it into $a_{n,m}^{\lessgtr}$ and giving the energy level $E_{n,m}^{\tau,s}$. The greater the index n is, the greater value of $a_{n,m}^{\lessgtr}$ will be, and one can easily relate this with different discrete energy levels from low to high in such QD system. The numeric result on the level $E_{0,m}^{\tau,s}$ with $\tau = -s$ and $m = 0, \pm 1$ is demonstrated below:

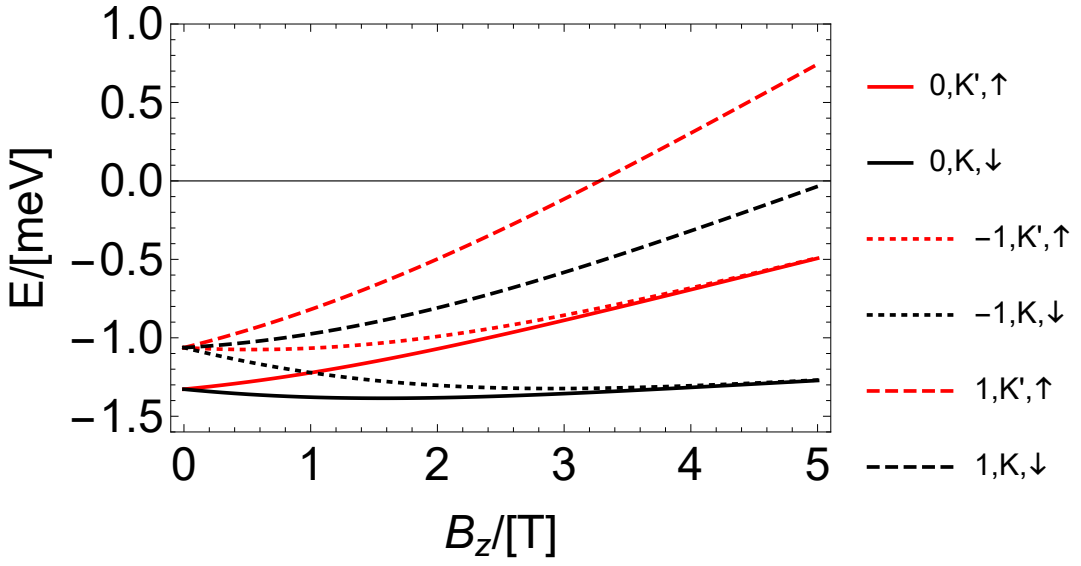


Figure 6: A few first energy levels ($n = 0$) of a MoS_2 QD with radius $R_d = 50\text{nm}$ depending on the magnetic field B_z.

One can see under low magnetic field, two lowest levels are $E_{0,0}^{\text{K},\downarrow}, E_{0,0}^{\text{K}',\uparrow}$. At higher field case, the levels with same τ, s are merging together. This phenomena will be explained in next section.

For the infinite high barrier limit $U_0 \to +\infty$, the solution of equation (24) will reduced to $u_m^{\tau,s}(\rho) = C^{<}M(a_m, |m|+1, \rho)\Theta(\rho_d - \rho)$. The continuous condition will simply be given by $M(a_m, |m|+$

$1, \rho_d) = 0$. Solving out a_m numerically, we are able to plot a series of energy level under magnetic field B_z.

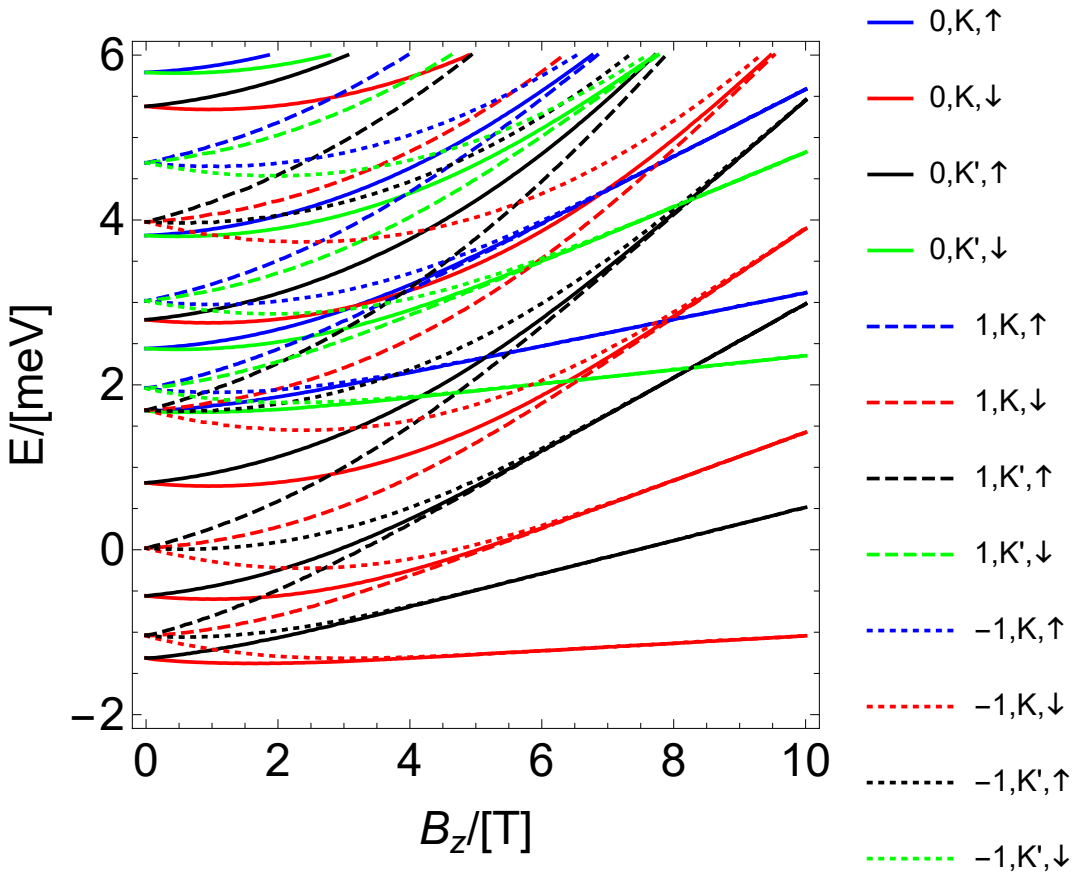


Figure 7: A more detail energy levels spectrum (higher n levels are presented) of the MoS_2 QD in dependence on the magnetic field B_z.

At low energy comparing with the finite barrier model, one can hardly tell difference between two numeric results. These two pictures are compatible. The levels merging informations are also in well agreement.

4.3 Landau levels

In QD system, we have derived and solved the Kummer's differential equation in inner- and outer region. From the numeric results we can tell the different QD levels will closely merge together bunch by bunch when magnetic field get large enough. These buchs of dense merging levels can be understood as an approaching to the normal degenerate Landau levels. To explain this, first we review the theory on the Landau level of free electrons in 2D plane. Applying previous notation the Hamiltonian of 2D electron in magnetic field can be derived as $H = \frac{1}{2m}(p_x^2 + p_y^2) + \frac{1}{2}m(\frac{\omega_c}{2})^2(x^2 + y^2) + \frac{\omega_c}{2}L_z$. With the help of algebraic approach,[20] one can reexpress the Hamiltonian with ladder operators $a^\dagger = (\frac{z}{2l_B} - l_B\partial_{\bar{z}})$, $a = (\frac{\bar{z}}{2l_B} + l_B\partial_z)$ and $b^\dagger = (\frac{\bar{z}}{2l_B} - l_B\partial_z)$, $b = (\frac{z}{2l_B} + l_B\partial_{\bar{z}})$ into $H = \frac{\hbar\omega_c}{2}(a^\dagger a + b^\dagger b + 1 + L_z/\hbar)$ where angular momentum operator L_z can be derived as $L_z = xp_y - yp_x = \hbar(a^\dagger a - b^\dagger b)$. Then the Landau levels show up as simple quantum harmonic oscillator levels $H = \hbar\omega_c(a^\dagger a + \frac{1}{2})$. It is easily known all the Landau levels wave functions can

be generated from lowest Landau level $\psi_{0,0}(z,\bar{z}) = \langle z,\bar{z}|N=0,M=0\rangle \sim e^{-|z|^2/4l_B^2}$ by applying ladder operator $a^\dagger$, which will rise the level energy quantum number $N \in \mathbb{N}$, and $b^\dagger$ rising another quantum number $M \in \mathbb{N}$ e.g. $(b^\dagger)^M|0,0\rangle = |0,M\rangle$, $\psi_{0,M} \sim \left(\frac{\bar{z}}{2l_B}\right)^M e^{-|z|^2/4l_B^2}$. One can check with angular momentum operator $L_z = \hbar(z\partial_z - \bar{z}\partial_{\bar{z}})$ that when acting on the lowest Landau level state $L_z\psi_{0,M} = -\hbar M\psi_{0,M}$ i.e. the angular momentum quantum number m takes the value $-M$. This quantum number M does not appear in the Landau level energy $E_{N,M} = (N+\frac{1}{2})\hbar\omega_c$, this means various M represents degenerate states for the same N. And we can see the lowest Landau level wave function ψ_{0M} with angular momentum M has $\langle r^2\rangle = 2l_B^2(M+1)$ implying that the peak of each wave function forms a circle with corresponding radius. Therefore in a planar disk with radius R, total number of the lowest Landau level states (Landau degeneracy) is given by $G = \frac{A}{\pi\Delta\langle r^2\rangle} = \frac{\pi R^2}{2\pi l_B^2} = \frac{\Phi}{\phi_0}$. For higher Landau level the picture is analog, however the angular momentum quantum number is now in general $m = N - M$ and the degeneracy of each Landau level remains the same.

Now we come back to the QD levels in TMDC, because in the large magnetic field case the magnetic length $l_B = \sqrt{\frac{\hbar}{eB_z}}$ is very small and the characteristic radius of those wave functions is hence relative small comparing to the QD radius. There can be more states with different angular momentum quantum number m squeezing in the circular QD region and sensing only quite small levels repulsive effect, which causes the energy splitting between different levels also known as quantum confining effect. This physical picture is therefore quite close to the normal Landau level mentioned above.

On the other aspect, if we adiabatically tune the potential barrier to zero, the model should strictly go back to free electron motion in TMDC under magnetic field, i.e. the Landau levels in TMDC. As we have previously discussed, the Kummer's equation has the general solution consisting first- and second kind confluent hypergeometric function, the continuous condition yields the Wronskian of them. An essential difference here is the paramter a is no longer piecewise in the whole region $[0,+\infty]$, therefore the condition reduces to $W\{M(a,|m|+1,\rho),U(a,|m|+1,\rho)\} = -\rho^{-(|m|+1)}e^\rho/\Gamma(a) = 0\,,\ \forall\rho\in[0,+\infty]$. The Wronskian will diverse at $+\infty$ unless $a = -n$ $(n\in\mathbb{N})$ then $\frac{1}{\Gamma(-n)} = 0$. The confluent hypergeometric functions in this case will reduce to associated Laguerre polynomials: $M(-n,|m|+1,\rho) = L_n^{|m|}(\rho)/C_n^{n+|m|}$, $U(-n,|m|+1,\rho) = (-1)^n n! L_n^{|m|}(\rho)$. This is in consistence with the differential equation: the Kummer's equation will also reduce to associated Laguerre's equation

$$\rho\frac{d^2}{d\rho^2}u_m^{\tau,s}(\rho) + (|m|+1-\rho)\frac{d}{d\rho}u_m^{\tau,s}(\rho) + nu_m^{\tau,s}(\rho) = 0. \tag{30}$$

The solution is naturally proportional to the associated Laguerre polynomial $L_n^{|m|}(\rho)$. On the other hand (25) shows that $\hbar\omega_c a = \frac{1}{2}(|m|+m+1-\tau)\hbar\omega_c + H^{\tau,s} - E$, we can calculate the energy level explicitly:

$$E_{n,m}^{\tau,s} = \Big(n + \frac{|m|+m}{2} + \frac{1-\tau}{2}\Big)\hbar\omega_c + H^{\tau,s}. \tag{31}$$

We can see the result is fully consistent with the free electron Landau level solved within polar coordinate in [20] and coincides with the result in [21]. Replacing m with $N-M$ one can find $n = M$ $(N > M)$, $n = N$ $(0 \le N < M)$, this also fits with the Landau level degeneracy structure analyzed before.

5 Disorder and spin-valley resonance

5.1 Disorder in monolayer MoS_2

In the momolayer TMDC material MoS_2, each primitive cell contain a metal atom M (Mo) and two chalcogen atoms X (S). Introducing atomic vacancy or doping another kind of atom as impurity, it can generate disorder in the lattice system. Typically a nonmagnetic defects/impurities of each type of atom place can simply take the form $V_i^{\mathrm{M(X)}}(\boldsymbol{r}) \otimes s_0$ where $V_i^{\mathrm{M(X)}}(\boldsymbol{r}) = V_i^{\mathrm{M(X)}}\delta(\boldsymbol{r} - \boldsymbol{R}_i)$.

Next we want to evaluate the matrix element of the disorder potential at low energy reflecting the scattering details. From the section of effective Hamiltonian, we know the valence band as well as conduction band in the lowest two bands model (8) are mainly constructed from $|d_{z^2},\boldsymbol{q},s\rangle \cong |\phi^{\mathrm{cb}}_{\boldsymbol{q},\tau,s}\rangle$ and $|d_{\tau 2},\boldsymbol{q},s\rangle = \frac{1}{\sqrt{2}}(|d_{x^2-y^2},\boldsymbol{q},s\rangle + \tau i|d_{xy},\boldsymbol{q},s\rangle) \cong |\phi^{\mathrm{vb}}_{\boldsymbol{q},\tau,s}\rangle$. We can transform these Bloch states from $\mathbf{k}$-space to real lattice space becoming Wannier states depending on the i-th primitive unit cell place vector $\boldsymbol{R}_i$ vice versa:

$$|\phi^{\mathrm{vb(cb)}}_{i,\tau,s}\rangle = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{q}} e^{-i\boldsymbol{q}\cdot\boldsymbol{R}_i}|\phi^{\mathrm{vb(cb)}}_{\boldsymbol{q},\tau,s}\rangle \tag{32}$$

$$|\phi^{\mathrm{vb(cb)}}_{\boldsymbol{q},\tau,s}\rangle = \frac{1}{\sqrt{N}}\sum_{i} e^{+i\boldsymbol{q}\cdot\boldsymbol{R}_i}|\phi^{\mathrm{vb(cb)}}_{i,\tau,s}\rangle, \tag{33}$$

Kaasbjerg *et al.*[22] has noticed that the on-site disorders in such 2D TMDC have C_3 symmetry, which means it is invariant under the threefold rotation by $\pm 2\pi/3$ around z-axis at defect place, i.e. $V_i^{\mathrm{M(X)}}(\boldsymbol{r}) = C_3 V_i^{\mathrm{M(X)}}(\boldsymbol{r})C_3^\dagger$. Since in TMDC space symmetry group is D_{3h}, at $\mathbf{K},\mathbf{K}'$ these high-symmetry points in the Brillouin zone the symmetry subgroup of the wave vector is $C_{3\mathrm{h}} = \{I, \sigma_h, C_3\}$. And it is straight forward the Bloch states at $\mathbf{K},\mathbf{K}'$ build up the irreducible representation of the subgroup. It is exactly the C_3 symmetry enforcing selection rules for the matrix elements between Bloch states:

$$\begin{aligned}
\langle\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}|V_i^{\mathrm{M(X)}}(\boldsymbol{r})|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau',s'}\rangle &= \langle\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}|C_3^\dagger C_3 V_i^{\mathrm{M(X)}}(\boldsymbol{r})C_3^\dagger C_3|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau',s'}\rangle \\
&= \langle\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}|C_3^\dagger V_i^{\mathrm{M(X)}}(\boldsymbol{r})C_3|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau',s'}\rangle \\
&= w^{\tau *}_{\mathrm{M(X)},\mathrm{vb(cb)}} w^{\tau'}_{\mathrm{M(X)},\mathrm{vb(cb)}}\langle\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}|V_i^{\mathrm{M(X)}}(\boldsymbol{r})|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau',s'}\rangle \\
&= \xi^{\tau\tau'}_{\mathrm{M(X)},\mathrm{vb(cb)}}\langle\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}|V_i^{\mathrm{M(X)}}(\boldsymbol{r})|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau',s'}\rangle,
\end{aligned} \tag{34}$$

where in the last two rows the C_3 were applied to the Bloch state and obtaining an extra phase factor $C_3|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}\rangle = w^{\tau}_{\mathrm{M(X)},\mathrm{vb(cb)}}|\phi^{\mathrm{vb(cb)}}_{\mathbf{K(K')},\tau,s}\rangle$ and $\xi^{\tau\tau'}_{\mathrm{M(X)},\mathrm{vb(cb)}} = w^{\tau *}_{\mathrm{M(X)},\mathrm{vb(cb)}} w^{\tau'}_{\mathrm{M(X)},\mathrm{vb(cb)}}$ is the total phase factor. Only when the phase factor is 1 then can the matrix element have non-zero value. Detail evaluation on the phase factor can be found in [22], here we only write out the result: when $\tau = \tau'$, $\xi^{\tau\tau'}_{\mathrm{M(X)},\mathrm{vb(cb)}} = 1$; when $\tau \neq \tau'$, we have

$$\begin{aligned}
\xi^{\tau\tau'}_{\mathrm{M,vb}} &= e^{\pm i8\pi/3} \quad , \quad \xi^{\tau\tau'}_{\mathrm{M,cb}} = 1 \\
\xi^{\tau\tau'}_{\mathrm{X,vb}} &= e^{\pm i4\pi/3} \quad , \quad \xi^{\tau\tau'}_{\mathrm{X,cb}} = e^{\pm i4\pi/3}
\end{aligned}$$

This implies only the M-centered defects can make non-zero contribution to intervalley ($\mathbf{K} \leftrightarrow \mathbf{K}'$) matrix element merely in the conduction band.

5.2 Valley Zeeman interaction induced by disorder

In our QD system, the quantum states are labeled as $|n, m; \tau, s\rangle$ with $n \in \mathbb{N}, m \in \mathbb{Z}, \tau = \mathrm{K}(\mathrm{K}'), s = \uparrow (\downarrow)$, the wave function is straight forward given by

$$\Psi^{\tau,s}_{n,m}(\rho, \varphi) = \langle \rho, \varphi | n, m; \tau, s\rangle = \frac{1}{\sqrt{2\pi}} e^{im\varphi} \rho^{|m|/2} e^{-\rho/2} u^{\tau,s}_{n,m}(\rho) \chi_s. \tag{35}$$

To investigate valley flipping process we focus on the lowest two level states $|0, 0; \mathrm{K}, \downarrow\rangle$, $|0, 0; \mathrm{K}', \uparrow\rangle$ at low magnetic field which are separated from higher energy level. Under low magnetic field, these two levels are quite close and the states are nearly a Kramer's pair spanning to a two dimensional Hilbert space. And it can be clearly noticed that the spinless or nonmagnetic disorders (defect/impurity) forbidden the scattering between these two states due to the fact that there is not any spin flipping process. Hence one way to resolve the problem is to introduce magnetic disorder mechanism, for example doping magnetic impurities to replace the original atoms in the lattice randomly. On the other hand as it has been discussed in last subsection, for the on-site disorder potential only the M-type can make valley flipping occur. Magnetic impurity will couple to the conduction electron, which is similar to the well studied s-d mixing in Kondo effect. Also after Schrieffer-Wolff perturbation one can effectively separate the charge- and spin fluctuation part, at low energy we only focus on spin fluctuation and will arrive at a Kondo type interaction term between conduction electron spin $\boldsymbol{s}$ and impurity spin $\boldsymbol{S}$ (for more details on the Kondo coupling one can refer to the book [18]). However we should note that here in the QD system we encounter only discrete QD level, therefore electron spin is constructed as $s_+ = |\uparrow\rangle\langle\downarrow|$, $s_- = |\downarrow\rangle\langle\uparrow|$, $s_z = |\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|$. Similarly the impurity spin can be defined with the impurity spin states. Hence we can effectively introduce a M-centered local spin coupling disorder term into the QD Hamiltonian:

$$H_{\mathrm{dis}} = \sum_i J_i \delta(\boldsymbol{r} - \boldsymbol{R}_i) \otimes (\boldsymbol{s} \cdot \boldsymbol{S}). \tag{36}$$

In addition we involve here only with the short-range disorder assumption indicating a zero mean value of the random on-site disorder potential $\langle J_i \rangle = 0$ (J_i is assumed to take value in the interval $[-\frac{J}{2}, \frac{J}{2}]$ randomly while $\langle\cdot\rangle$ means statistical average or mean according to the random variable distribution namely the disorder configuration), and the short-range property means spatial uncorrelated $\langle J_i J_j \rangle = \langle J_i^2 \rangle \delta_{ij}$.

The goal here is to obtain a valley mixing term from the disorder term. Now we temporally omit the n, m quantum number and write the two QD states into tensor product of valley part and spin part $|\mathrm{K}\rangle \otimes |\downarrow\rangle$, $|\mathrm{K}'\rangle \otimes |\uparrow\rangle$.These two states of interest span our Hilbert space and the Hamiltonian can be in principle expressed with the projection operator owing to completeness, which has been similarly done before on the carbon nanotube model.[23] The projection operator is written as

$$\begin{aligned} P &= |\mathrm{K}, \downarrow\rangle\langle \mathrm{K}, \downarrow| + |\mathrm{K}', \uparrow\rangle\langle \mathrm{K}', \uparrow| \\ &= |\mathrm{K}\rangle\langle \mathrm{K}| \otimes |\downarrow\rangle\langle\downarrow| + |\mathrm{K}'\rangle\langle \mathrm{K}'| \otimes |\uparrow\rangle\langle\uparrow| = P_{\mathrm{K}} \otimes P_{\downarrow} + P_{\mathrm{K}'} \otimes P_{\uparrow}. \end{aligned} \tag{37}$$

Then the disorder term can be expressed in this projective representation as:

$$P H_{\mathrm{dis}} P = P(H^{\mathrm{orbit}}_{\mathrm{dis}} \otimes H^{\mathrm{spin}}_{\mathrm{dis}}) P = \sum_{\tau,\tau';s,s'} (P_\tau H^{\mathrm{orbit}}_{\mathrm{dis}} P_{\tau'}) \otimes (P_s H^{\mathrm{spin}}_{\mathrm{dis}} P_{s'}) \delta_{s,-\tau} \delta_{s',-\tau'}. \tag{38}$$

The orbital part is to be investigated first. We can formally write down this part under Pauli matrix basis in valley space $\{\tau_0, \tau_x, \tau_y, \tau_z\}$:

$$\sum_{\tau,\tau'} P_\tau H^{\mathrm{orbit}}_{\mathrm{dis}} P_{\tau'} = b_0 \tau_0 + \boldsymbol{b} \cdot \boldsymbol{\tau}, \tag{39}$$

where the parameters are given by:

$$b_0 = \sum_i [\langle \mathrm{K}|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle + \langle \mathrm{K}'|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}'\rangle] \tag{40}$$

$$b_x = \sum_i [\langle \mathrm{K}|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}'\rangle + \langle \mathrm{K}'|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle] \tag{41}$$

$$b_y = \sum_i [i\langle \mathrm{K}|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}'\rangle - i\langle \mathrm{K}'|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle] \tag{42}$$

$$b_z = \sum_i [\langle \mathrm{K}|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle - \langle \mathrm{K}'|J_i\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}'\rangle]. \tag{43}$$

The first term with b_0 is merely shifting the energy level hence we can drop it out by redefining the energy level value. The last term with b_z is relative small and can be omitted. This can be justified with the such way: we know the two level are Kramer's pair when $B_z = 0$ and with time-reversal symmetry $\mathcal{T}|\mathrm{K},\downarrow\rangle = |\mathrm{K}',\uparrow\rangle$ it can be proven easily that $b_z = 0$; when perpendicular magnetic field is present however at low field situation, these two states are nearly Kramer's pair (nearly degenerate) both with great magnetic length much larger then the QD radius $l_B \gg R_d$, therefore b_z should only slightly deviate from zero. The remaining terms are $b_x\tau_x$ and $b_y\tau_y$ and we now have obtained an effective valley-Zeeman interaction in x-y plane:

$$\begin{aligned} H_Z^v = b_x\tau_x + by\tau_y &= \begin{pmatrix} 0 & b_x - ib_y \\ b_x + ib_y & 0 \end{pmatrix} \\ &= b\begin{pmatrix} 0 & e^{-i\delta} \\ e^{i\delta} & 0 \end{pmatrix} = be^{i\delta}\tau_- + be^{-i\delta}\tau_+ \\ &= be^{i\delta}|\mathrm{K}'\rangle\langle \mathrm{K}| + \mathrm{h.c.}, \end{aligned} \tag{44}$$

in above the amplitude $b = \sqrt{b_x^2 + b_y^2} = 2\sum_i J_i|\langle \mathrm{K}'|\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle|$ and the angle $\delta = \arctan(b_y/b_x) = \arctan[\mathrm{Im}\langle \mathrm{K}'|\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle/\mathrm{Re}\langle \mathrm{K}'|\delta(\boldsymbol{r}-\boldsymbol{R}_i)|\mathrm{K}\rangle]$.

At last we consider further the spin part $\sum_{s,s'} P_s(\boldsymbol{s}\cdot\boldsymbol{S})P_{s'}$ and it can be easily figured out the spin-valley Zeeman interaction finally reads:

$$H_Z^{s\text{-}v} = be^{i\delta}(\tau_- s_+)\otimes S^- + \mathrm{h.c.} = be^{i\delta}|\mathrm{K}',\uparrow\rangle\langle \mathrm{K},\downarrow|\otimes S^- + \mathrm{h.c.}. \tag{45}$$

5.3 Spin-valley resonance

In order to drive the electron valley resonance, we apply a AC voltage on the gates along x-axis causing a periodic dispalcement on the confining potential barrier $U(\boldsymbol{r}) \to U(\boldsymbol{r},t) = U(\boldsymbol{r} - X\sin\omega t\,\mathbf{e}_x)$ with AC frequency $\omega/2\pi$. We then transform the system to the co-moving frame for convenience with unitary operator $\mathcal{U}(t) = e^{\frac{i}{\hbar}X\sin(\omega t)p_x}$ which is exactly the displacement operator. The quantum state of this system in co-moving frame will then turns back to original one in rest QD system $\mathcal{U}(t)|\widetilde{\Psi}(t)\rangle = |\Psi(t)\rangle$, the Schrödinger equation will then transformed into:

$$\begin{aligned} & i\hbar\partial_t|\widetilde{\Psi}(t)\rangle = [H_0 + U(\boldsymbol{r},t) + H_{\mathrm{dis}}(\boldsymbol{r})]|\widetilde{\Psi}(t)\rangle \\ \Rightarrow & i\hbar\partial_t(\mathcal{U}^\dagger(t)|\Psi(t)\rangle) = [H_0 + U(\boldsymbol{r},t) + H_{\mathrm{dis}}(\boldsymbol{r})]\mathcal{U}^\dagger(t)|\Psi(t)\rangle \\ \Rightarrow & \mathcal{U}^\dagger(t)i\hbar\partial_t|\Psi(t)\rangle + i\hbar\dot{\mathcal{U}}^\dagger(t)|\Psi(t)\rangle = [H_0 + U(\boldsymbol{r},t) + H_{\mathrm{dis}}(\boldsymbol{r})]\mathcal{U}^\dagger(t)|\Psi(t)\rangle \\ \Rightarrow & i\hbar\partial_t|\Psi(t)\rangle = [i\hbar\dot{\mathcal{U}}(t)\mathcal{U}^\dagger(t) + H_0 + U(\boldsymbol{r}) + \mathcal{U}(t)H_{\mathrm{dis}}(\boldsymbol{r})\mathcal{U}^\dagger(t)]|\Psi(t)\rangle. \end{aligned}$$

We can see in co-moving frame the disorder term become dynamic $\mathcal{U}(t)H_{\mathrm{dis}}(\boldsymbol{r})\mathcal{U}^\dagger(t) = H_{\mathrm{dis}}(\boldsymbol{r} + X\sin\omega t\,\mathbf{e}_x)$. There is still an additional time evolving term $i\hbar\dot{\mathcal{U}}(t)\mathcal{U}^\dagger(t)$ causing QD level slightly altering periodically, since it does not mix two levels we can neglect it here.

Next we would like to investigate how the spin-valley Zeeman interaction will deform. Assuming the oscillating displacement is small comparing to magnetic length $X/l_B \ll 1$, then we expand the unitary displacement operator to the first order $\mathcal{U}(t) \approx 1 + \frac{i}{\hbar} X \sin\omega t\, p_x = 1 + X \sin\omega t\, \partial_x$. The Zeeman field parameter $be^{i\delta}$ can then be evaluated with $2\sum_i J_i \langle \mathrm{K}'|\delta(\boldsymbol{r} - \boldsymbol{R}_i + X\sin\omega t\, \mathbf{e}_x)|\mathrm{K}\rangle = 2\sum_i J_i \langle \mathrm{K}'|\mathcal{U}(t)\delta(\boldsymbol{r} - \boldsymbol{R}_i)\mathcal{U}^\dagger(t)|\mathrm{K}\rangle$. Inserting the completeness identity $1 = \int d\boldsymbol{r}|\boldsymbol{r}\rangle\langle\boldsymbol{r}|$ in two the matrix element the wave function can finally be brought in the calculation:

$$\begin{aligned} be^{i\delta} &= 2\sum_i J_i \langle \mathrm{K}'|\mathcal{U}(t)\delta(\boldsymbol{r} - \boldsymbol{R}_i)\mathcal{U}^\dagger(t)|\mathrm{K}\rangle \\ &= 2\sum_i J_i \int_0^{2\pi} d\varphi \int_0^{+\infty} d\rho\, [\mathcal{U}^\dagger(t)\Psi^{\mathrm{K}'}_{0,0}(\rho,\varphi)]^* \delta(\rho-\rho_i)\delta(\varphi-\varphi_i)[\mathcal{U}^\dagger(t)\Psi^{\mathrm{K}}_{0,0}(\rho,\varphi)] \\ &= 2\sum_i J_i [\mathcal{U}^\dagger(t)\Psi^{\mathrm{K}'}_{0,0}(\rho,\varphi)]^*_{\rho_i,\varphi_i} [\mathcal{U}^\dagger(t)\Psi^{\mathrm{K}}_{0,0}(\rho,\varphi)]_{\rho_i,\varphi_i} \end{aligned} \tag{46}$$

Applying the approximate form of $\mathcal{U}(t)$ and $\Psi^\tau_{0,0}(\rho,\varphi)$ we can calculate

$$\begin{aligned} \mathcal{U}^\dagger(t)\Psi^\tau_{0,0}(\rho,\varphi) &= (1 - X\sin\omega t\,\partial_x)\Psi^\tau_{0,0}(\rho,\varphi) \\ &= \Big[1 - X\sin\omega t\Big(\cos\varphi\partial_r - \frac{1}{r}\sin\varphi\partial_\varphi\Big)\Big]\Big(\frac{1}{\sqrt{2\pi}} e^{-\rho/2} u^\tau_{0,0}(\rho)\Big) \\ &= \Psi^\tau_{0,0}(\rho,\varphi) - \frac{X}{l_B}\sin\omega t \cdot \frac{1}{\sqrt{2\pi}}\cos\varphi\sqrt{2\rho}\partial_\rho[e^{-\rho/2}u^\tau_{0,0}(\rho)]. \end{aligned} \tag{47}$$

Substituting back to (46) and keep the term only till the first order of X/l_B, we get:

$$\begin{aligned} be^{i\delta} &\approx b_1 e^{i\delta_1} + b_2 e^{i\delta_2}\sin\omega t \quad (\delta_1 = \delta_2 = 0) \\ &= 2\sum_i \frac{J_i}{2\pi} e^{-\rho_i} u^{\mathrm{K}'}_{0,0}(\rho_i) u^{\mathrm{K}}_{0,0}(\rho_i) \\ &\quad - \frac{X}{l_B}\sin\omega t \cdot 4\sum_i \frac{J_i}{\pi}\cos\varphi_i\ \rho_i \partial_\rho[e^{-\rho_i/2}u^{\mathrm{K}'}_{0,0}(\rho_i)]\partial_\rho[e^{-\rho_i/2}u^{\mathrm{K}}_{0,0}(\rho_i)]. \end{aligned} \tag{48}$$

This disorder induced spin-valley Zeeman field $\boldsymbol{b} = (b_x, b_y, 0)$ is randomly fluctuating in lattice space because it depends on the disorder confuguration. For the short-range disorder we known $\langle J_i\rangle = 0$ causing $\langle b_{1(2)}\rangle = \sum_i \langle J_i\rangle F_{1(2)}(\rho_i,\varphi_i) = 0$ while $\langle b_1^2\rangle \propto \langle b_2^2\rangle \neq 0$ due to $\langle J_i J_j\rangle = \langle J_i^2\rangle\delta_{ij}$. Summing the primitive unit cell over the full lattice is equivalent to perform continuous integral, we calculate $\langle b^2_{1(2)}\rangle$ with integral evaluation:

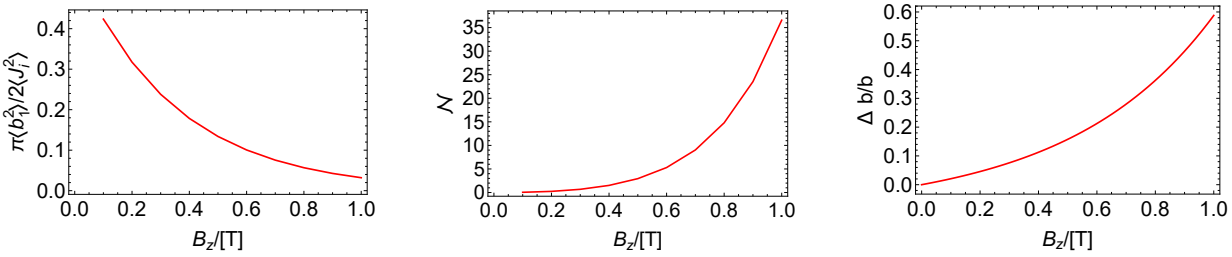
$$\begin{aligned} \langle b_1^2\rangle &= \int_0^{2\pi} d\varphi \int_0^{+\infty} d\rho\ \frac{\langle J_i^2\rangle}{\pi^2} e^{-2\rho}[u^{\mathrm{K}'}_{0,0}(\rho)u^{\mathrm{K}}_{0,0}(\rho)]^2 \\ &= \frac{2\langle J_i^2\rangle}{\pi}\int_0^{+\infty} d\rho\ e^{-2\rho}[u^{\mathrm{K}'}_{0,0}(\rho)u^{\mathrm{K}}_{0,0}(\rho)]^2 \\ &= \frac{2\langle J_i^2\rangle}{\pi}\int_0^{+\infty} d\rho\ e^{-2\rho}[u^{\mathrm{K}}_{0,0}(\rho)]^4 \end{aligned} \tag{49}$$

$$\begin{aligned} \langle b_2^2\rangle &= \Big(\frac{X}{l_B}\Big)^2 \int_0^{2\pi} d\varphi \int_0^{+\infty} d\rho\ 16\frac{\langle J_i^2\rangle}{\pi^2}\cos^2\varphi\ \rho^2\{\partial_\rho[e^{-\rho/2}u^{\mathrm{K}'}_{0,0}(\rho)]\partial_\rho[e^{-\rho/2}u^{\mathrm{K}}_{0,0}(\rho)]\}^2 \\ &= \Big(\frac{X}{l_B}\Big)^2 \frac{16\langle J_i^2\rangle}{\pi}\int_0^{+\infty} d\rho\ \rho^2\{\partial_\rho[e^{-\rho/2}u^{\mathrm{K}'}_{0,0}(\rho)]\partial_\rho[e^{-\rho/2}u^{\mathrm{K}}_{0,0}(\rho)]\}^2 \\ &= \Big(\frac{X}{l_B}\Big)^2 \frac{16\langle J_i^2\rangle}{\pi}\int_0^{+\infty} d\rho\ \rho^2\{\partial_\rho[e^{-\rho/2}u^{\mathrm{K}}_{0,0}(\rho)]\}^4 \end{aligned} \tag{50}$$

In above derivation we have used the fact $u^{\mathrm{K}}_{n,m}(\rho) = u^{\mathrm{K}'}_{n,m}(\rho)$ for the confluent hypergeometric functions with same n, m the parameter $a_{n,m}$ is the same no matter what the valley index is. We

evaluate the integral in (49) with $u^{\mathrm{K}}_{0,0}(\rho) = C^{<}M(a,1,\rho)\Theta(\rho_d - \rho)$ under different B_z numerically and fit the data with a nonlinear curve $e^{k_1 B_z + k_2}$. The result is shown in Fig.8 (a) and fitting coefficient is $k_1 = -2.87 \pm 0.002$, $k_2 = -0.57 \pm 0.0005$.

To observe the relation between $\langle b_1^2\rangle$ and $\langle b_2^2\rangle$, we start with an example when $B_z = 0.5\mathrm{T}$, the integrals are numerically calculated and we find $\langle b_2^2\rangle \approx 2.94(\frac{X}{l_B})^2\langle b_1^2\rangle$. More generally we can evaluate the rate $\mathcal{N} = \frac{\langle b_2^2\rangle}{\langle b_1^2\rangle}/(\frac{X}{l_B})^2$ under various B_z, the Fig.8 shows that the rate $\mathcal{N}$ increase rapidly during the magnetic field increasing. This is in principle positive to our requirement on obtaining strong spin-valley Zeeman field and now it can be achieved by simply rising the magnetic field, however higher magnetic field will also bring in the level crossing from $m \neq 0$ states which we want to avoid. According to the energy level diagram shown in Fig.6 , a proper interval of magnetic field for investigation will be $0 < B_z < 1\mathrm{T}$. Taking $X = 2.5\mathrm{nm}$ we get $\langle b_2^2\rangle \approx \mathcal{N}B_z(\frac{2.5\mathrm{nm}}{25\mathrm{nm}})^2\langle b_1^2\rangle = 0.01\mathcal{N}B_z\langle b_1^2\rangle$. Similarly using a nonlinear curve fitting, we can numerically suggest an approximate relation between $\mathcal{N}$ and B_z: $\mathcal{N}(B_z) = e^{k_3 B_z} - 1$ where the value of fitting coefficient is $k_3 = 3.57 \pm 0.04$. Then we now denote $b = \sqrt{\langle b_1^2\rangle}$ and $\Delta b = b\sqrt{0.01\mathcal{N}(B_z)\cdot B_z} = \sqrt{\langle b_2^2\rangle}$.



(a) Integral from $\langle b_1^2\rangle$ (b) $\mathcal{N}(B_z)$ (c) $\Delta b(B_z)/b$

Figure 8: The plotting of the numeric results of the integral part of $\langle b_1^2\rangle$, $\mathcal{N}$ and $\Delta b/b$ as a function of magnetic field B_z.

We have already known the energy of the two level $E^{\mathrm{K}\downarrow}_{0,0}, E^{\mathrm{K}'\uparrow}_{0,0}$, we now employ another notation for the levels and states: $E^{\mathrm{K}\downarrow}_{0,0} \to E_1$, $E^{\mathrm{K}'\uparrow}_{0,0} \to E_2$ and $|0,0;\mathrm{K},\downarrow\rangle \to |1\rangle$, $|0,0;\mathrm{K}',\uparrow\rangle \to |2\rangle$. Strictly speaking the Hilbert space of our system is the tensor space from QD two-level system and impurity spin i.e. $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_{2\mathrm{lv}} \otimes \mathcal{H}_S$, the total state is a product state $|\Psi_{\mathrm{tot}}(t)\rangle = |\psi_{2\mathrm{lv}}(t)\rangle \otimes |S\rangle = U(t)|\psi_{2\mathrm{lv}}(0)\rangle \otimes |S\rangle$ where $U(t) = \mathbf{T}e^{-\frac{i}{\hbar}\int_0^t H(t')dt'}$ is the time evolution operator. Since the impurity spin energy are not in our interest, we can effectively wipe out the impurity spin operator from the term (45) by tracing out the impurity spin in the total density matrix (namely partial trace) $\mathrm{Tr}_S\hat{\rho}_{\mathrm{tot}}(t) = \sum_{S_z}\langle S_z|\hat{\rho}_{\mathrm{tot}}(t)|S_z\rangle = \hat{\rho}_{2\mathrm{lv}}(t)$ and turning back to the Hamiltonian it is equivalent to merely take away S^-, S^+ without changing the rest. Defining the energy difference with $\hbar\omega_0 = E_2 - E_1 = E^{\mathrm{K}'\uparrow}_{0,0} - E^{\mathrm{K}\downarrow}_{0,0} = \hbar\omega_c$ and the mean energy $\bar{\varepsilon} = (E_1 + E_2)/2$, we can rearrange the energy for two state as $E_1 \to -\hbar\omega_0/2$, $E_2 \to +\hbar\omega_0/2$ and drop out the mean energy $\bar{\varepsilon}$. Again applying the Pauli matrix representation $\sigma^z = |1\rangle\langle 1| - |2\rangle\langle 2|$, $\sigma^x = |1\rangle\langle 2| + |2\rangle\langle 1|$, the Hamiltonian for the total two-level system now reads:

$$H_{\mathrm{Rabi}} = -\frac{\hbar\omega_0}{2}\sigma^z + (b - \Delta b\sin\omega t)\sigma^x. \tag{51}$$

It describes a Rabi type oscillation process, which is exactly the desire spin-valley resonance process. For the basic knowledge on Rabi oscillation we recommend the book [24]. To solve this time evolution system, the quantum state is supposed to take the form:

$$|\psi(t)\rangle = c_1(t)e^{-iE_1t/\hbar}|1\rangle + c_2(t)e^{-iE_2t/\hbar}|2\rangle, \tag{52}$$

and substituting into time-dependent Schrödinger equation $i\hbar\partial_t|\psi(t)\rangle = H|\psi(t)\rangle$ we get the equation of motion for $c_1(t), c_2(t)$ as follow

$$\dot{c}_1(t) = -\frac{i}{\hbar}[b - \Delta b \sin\omega t]e^{-i\omega_0 t}c_2(t) \tag{53}$$

$$\dot{c}_2(t) = -\frac{i}{\hbar}[b - \Delta b \sin\omega t]e^{i\omega_0 t}c_1(t). \tag{54}$$

With the help of Euler's formula, we have $\sin\omega t = \frac{e^{i\omega t}-e^{-i\omega t}}{2}$ and combining with $e^{\pm i\omega_0 t}$ surely we will get $e^{\pm i(\omega-\omega_0)t}$ part as well as $e^{\pm i(\omega+\omega_0)t}$ in the equations. This second part is fast rotation factor and after time average it can be self-canceled therefore we approximately set $e^{\pm i(\omega+\omega_0)t} \to 0$. Similarly comparing to $e^{\pm i(\omega-\omega_0)t}$ the $e^{\pm i\omega_0 t}$ is also relative fast due to the fact that the applied AC frequency is usually tuned very closed to ω_0 to achieve resonance, hence this factor $e^{\pm i\omega_0 t}$ in the equation should also be neglected. Such kind of approximation is the so called "Rotational Wave Approximation". The equations of motion then become

$$\dot{c}_1(t) = \frac{\Delta b}{2\hbar}e^{i(\omega-\omega_0)t}c_2(t) \tag{55}$$

$$\dot{c}_2(t) = -\frac{\Delta b}{2\hbar}e^{-i(\omega-\omega_0)t}c_1(t). \tag{56}$$

Further we attempt to eliminate $c_1(t)$ so that we obtain a pure second order differential equation for $c_2(t)$:

$$\ddot{c}_2(t) + i(\omega-\omega_0)\dot{c}_2(t) + \frac{(\Delta b)^2}{4\hbar^2}c_2(t) = 0 \tag{57}$$

Solving this normal second order differential with $c_2(t) = e^{i\lambda t}$ and one can obtain an equation $\lambda^2 + \Delta\omega\lambda - \frac{(\Delta b)^2}{4\hbar^2} = 0$ and find two roots:

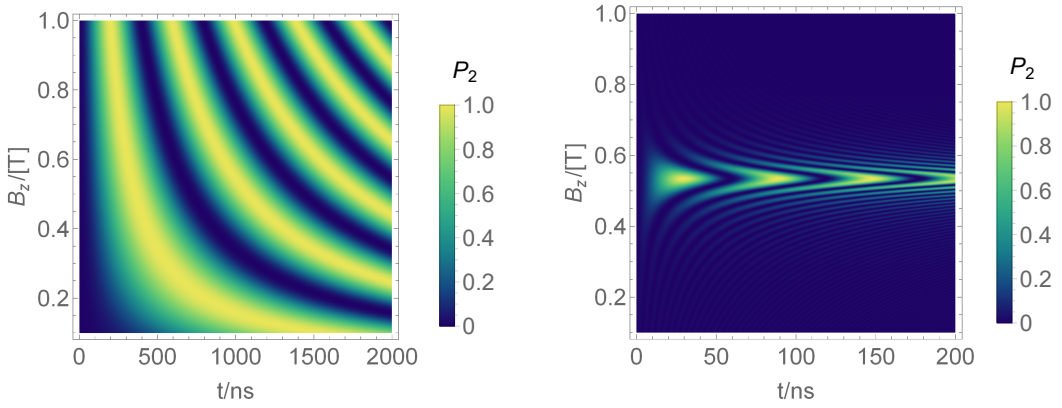
$$\lambda_\pm = \frac{1}{2}\Big[-\Delta\omega \pm \sqrt{(\Delta\omega)^2 + (\Delta b/\hbar)^2}\Big]. \tag{58}$$

The general solution is the linear combination of these two roots solutions $c_- e^{i\lambda_- t} + c_+ e^{i\lambda_+ t}$, assuming the system start from level-1 namely the ground state, which implies the initial conditions $c_1(0) = 1\,(\dot{c}_2(0) = -\Delta b/2\hbar)$, $c_2(0) = 0\ (\dot{c}_1(0) = 0)$. Finally the explicit form of these two time-evolving coefficients can be derived as

$$c_1(t) = e^{i\Delta\omega t/2}\Big[\cos(\omega_{\rm R}t/2) - i\frac{\Delta\omega}{\omega_{\rm R}}\sin(\omega_{\rm R}t/2)\Big] \tag{59}$$

$$c_2(t) = -\frac{\Delta b}{\hbar\omega_{\rm R}}e^{-i\Delta\omega t/2}\sin(\omega_{\rm R}t/2), \tag{60}$$

where $\omega_{\rm R} = \sqrt{(\Delta\omega)^2 + (\Delta b/\hbar)^2}$ is the Rabi frequency. The population on the excited state (level-2) is given by $P_2(t) = |c_2(t)|^2 = \frac{(\Delta b)^2}{\hbar^2\omega_{\rm R}^2}\sin^2(\frac{\omega_{\rm R}t}{2})$ and $(\Delta b)^2 = \langle b_2^2\rangle$ relies on B_z. When the detuning $\Delta\omega$ is zero, the system reaches the Rabi resonance and $P_1(t) + P_2(t) = 1$, hence our spin-valley resonance frequency here is directly the cyclotron frequency in TMDC. The population becomes $P_2(t) = \sin^2(\frac{\Delta b t}{2\hbar})$. Analogously for the normal finite detuning situation, still by employing the "Rotational Wave Approximation" with an extra approximation $(\Delta b/b)^2 \ll 1$, the last term in the differential equation (57) will simply change to $b^2/\hbar^2$. We will obtain an alternative Rabi frequency $\omega_{\rm R} = \sqrt{(\Delta\omega)^2 + 4(b/\hbar)^2}$ and the population will become $P_2(t) = \frac{4b^2}{\hbar^2\omega_R^2}\sin^2(\frac{\omega_R t}{2}) < 1$. Setting the parameter $\sqrt{2\langle J_i^2\rangle/\pi} = 0.01$ the population of level-2 in various status are plotted in the Fig.9 below. From the diagram we can see with the fixed AC driving frequency the oscillation frequency is low when the detuning is getting to vanish while the maximum of population is close to 1. In other magnetic field region oscillation frequency is extremely large however the maximum of population decreases very steeply away from zero detuning therefore is very small. It implies that these fast oscillation with larger detuning should be hard to be observed in the experiment.



(a) $\Delta\omega = 0$ (b) $\omega = 2\,\mathrm{GHz}$

Figure 9: Population pattern of QD level-2 evolving periodically along time depending on magnetic field B_z, the brightness resembles the value of population. (a) Zero detuning $\Delta\omega = 0$, i.e. resonance case. (b) Finite detuning case with constant AC driving frequency $\omega = 2\,\mathrm{GHz}$.

5.4 Quantum gate operation

It is well known that by rotating into the rotating frame, the Rabi model can be mapped to a quantum gate model.[24] The rotation operator is written as $\mathcal{R}(t) = e^{-i\frac{\omega_0 t}{2}\sigma^z}$, similar to the previous treatment one should expect the Hamiltonian in the rotating frame has the form $H_{\mathrm{rot}} = \mathcal{R}(t)H\mathcal{R}^\dagger(t) + i\hbar\dot{\mathcal{R}}(t)\mathcal{R}^\dagger(t)$. For the Hamiltonian (51), applying $e^{-i\frac{\omega_0 t}{2}\sigma^z} = \mathbb{I}\cos(\frac{\omega_0 t}{2}) - i\sigma^z\sin(\frac{\omega_0 t}{2})$ we derive the Hamiltonian as follow,

$$
\begin{aligned}
H_{\mathrm{rot}} &= -\frac{\hbar\omega_0}{2}e^{-i\frac{\omega_0 t}{2}\sigma^z}\sigma^z e^{i\frac{\omega_0 t}{2}\sigma^z} + (b - \Delta b\sin\omega t)e^{-i\frac{\omega_0 t}{2}\sigma^z}\sigma^x e^{i\frac{\omega_0 t}{2}\sigma^z} + i\hbar(\partial_t e^{-i\frac{\omega_0 t}{2}\sigma^z})e^{i\frac{\omega_0 t}{2}\sigma^z} \\
&= -\frac{\hbar\omega_0}{2}\sigma^z + (b - \Delta b\sin\omega t)\begin{pmatrix} e^{-i\frac{\omega_0 t}{2}} & 0 \\ 0 & e^{i\frac{\omega_0 t}{2}} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{i\frac{\omega_0 t}{2}} & 01 \\ 0 & e^{-i\frac{\omega_0 t}{2}} \end{pmatrix} + \frac{\hbar\omega_0}{2}\sigma^z \\
&= b\begin{pmatrix} 0 & e^{-i\omega_0 t} \\ e^{i\omega_0 t} & 0 \end{pmatrix} - \frac{\Delta b}{2}\begin{pmatrix} 0 & e^{i(\omega-\omega_0)t} \\ e^{i(\omega+\omega_0)t} & 0 \end{pmatrix} - \frac{\Delta b}{2}\begin{pmatrix} 0 & e^{-i(\omega+\omega_0)t} \\ e^{-i(\omega-\omega_0)t} & 0 \end{pmatrix}.
\end{aligned}
$$

Employing "Rotation Wave Approximation" again we neglect $e^{\pm i(\omega+\omega_0)t}$, the rotating frame Hamiltonian reads

$$
\begin{aligned}
H_{\mathrm{rot}} &= b\begin{pmatrix} 0 & e^{-i\omega_0 t} \\ e^{i\omega_0 t} & 0 \end{pmatrix} - \frac{\Delta b}{2}\begin{pmatrix} 0 & e^{i\Delta\omega t} \\ e^{-i\Delta\omega t} & 0 \end{pmatrix} \\
&= (b\cos\omega_0 t - \Delta b\cos\Delta\omega t)\sigma^x + (b\sin\omega_0 t + \Delta b\sin\Delta\omega t)\sigma^y. \qquad (61)
\end{aligned}
$$

At the resonant status ($\Delta\omega = 0$) the gate is self-rotating with the frequency $\omega_0 = \omega_c$. Under low magnetic field this slow rotation can be omitted, and one can get a σ^x gate. Performing AC electric field causing detuning within certain time period allows us to obtain various gate combing with σ^x and σ^y so as to carry out quantum manipulation when dealing with quantum information and computation process.

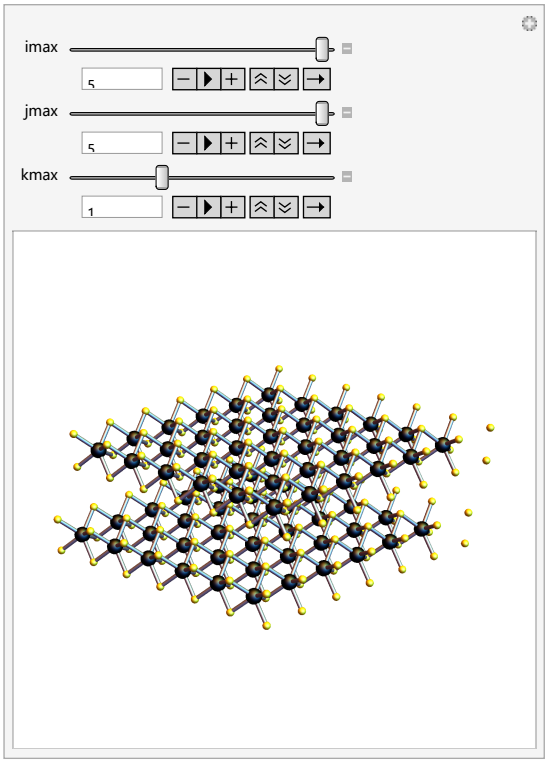
6 Conclusion

In this thesis we have investigated the spin-valley resonance process in quantum dot (QD) system constructed in monolayer transition metal dichalcogenides material (TMDCs) mainly MoS_2 under magnetic field. We have chosen the tight-biding model as the starting point to describe energy bands structure in TMDCs. Using Schrieffer-Wolff transformation we have got effective one-band model, with suitable parameters value the feature band structure near K-points can be recovered. By figuring out the QD energy levels under the magnetic field we find a clear crossover on the between discrete levels due to quantum confinement and the Landau levels convincing us the correct physics is founded within this research. Further we establish the key model of spin-valley resonance with spin-valley Zeeman field mediated by disorder mechanism from magnetic impurities, which enables the spin- and valley flipping process simultaneously. Then projecting the model with respect to the two lowest nearly Kramer's pair QD states under low magnetic field, we obtain a Rabi cycle like model which describes the main physics picture of the spin-valley resonance. As it can be seen, the detuning of the oscillation process will suppress the intensity of population however increase sharply the frequency. This result provide us useful information on the operation with spin-valley resonance in TMDCs. In addition we have a brief look at the application of our model into the quantum gate. By switching on proper driving electric AC field on the system one can obtain different quantum gate after certain working time. We believe all these results has a positive contribution to the studying on the field quantum information processing and two dimensional semiconductor as well.

Appendix

The code for drawing lattice structure of TMDCs is demonstrated below. In this work the code was rendered in program *Mathematica* version 10 and the credit of development is owed to the user "bobthechemist" from Internet forum *Mathematica*®**StackExchange**. The original resource can be found in the post: https://mathematica.stackexchange.com/questions/61249/constructing-primitive-cell-using-information-from-chemicaldata.

```
hcp=LatticeData["HexagonalClosePacking","Image"];
spheres=Cases[hcp[[1]],Sphere[x_,y_]:>x,Infinity];
unitcellspheres=spheres[[{1,2,4,5,11,12,14,15}]];
unitcellspheres[[All,3]]=unitcellspheres[[All,3]]/.{1->0.5,-1->-0.5};
csphere={-1/2,1/(2 Sqrt[3]),0};
unitcell2=GraphicsComplex[Join[unitcellspheres,{csphere}],
{(*'global' directives*)Specularity[GrayLevel[1],9],
(*sulfur atoms*){Yellow,Table[Sphere[i,0.1],{i,1,8}]},
(*Mo*){Black,Sphere[9,0.2]},
(*Mo bonds*){GrayLevel[0.5],
Tube[{#,9},0.05]&/@{1,2,3,5,6,7}}}]//Graphics3D;
Manipulate[Graphics3D[
GeometricTransformation[unitcell2[[1]],
Flatten@{TranslationTransform[{0,0,0}],
Table[TranslationTransform[{j 0.5+i,j Sqrt[3]/2+Mod[k,2],2 k}],
{i,0,imax},{j,0,jmax},{k,0,kmax}]}],
SphericalRegion->True,Boxed->False],
{imax,0,5,1},{jmax,0,5,1},{kmax,0,3,1}]
```



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